August 14, 2025

Journal Club:

REC-CAGEFREE & AGENT IDE Trials

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Inova™

Agenda

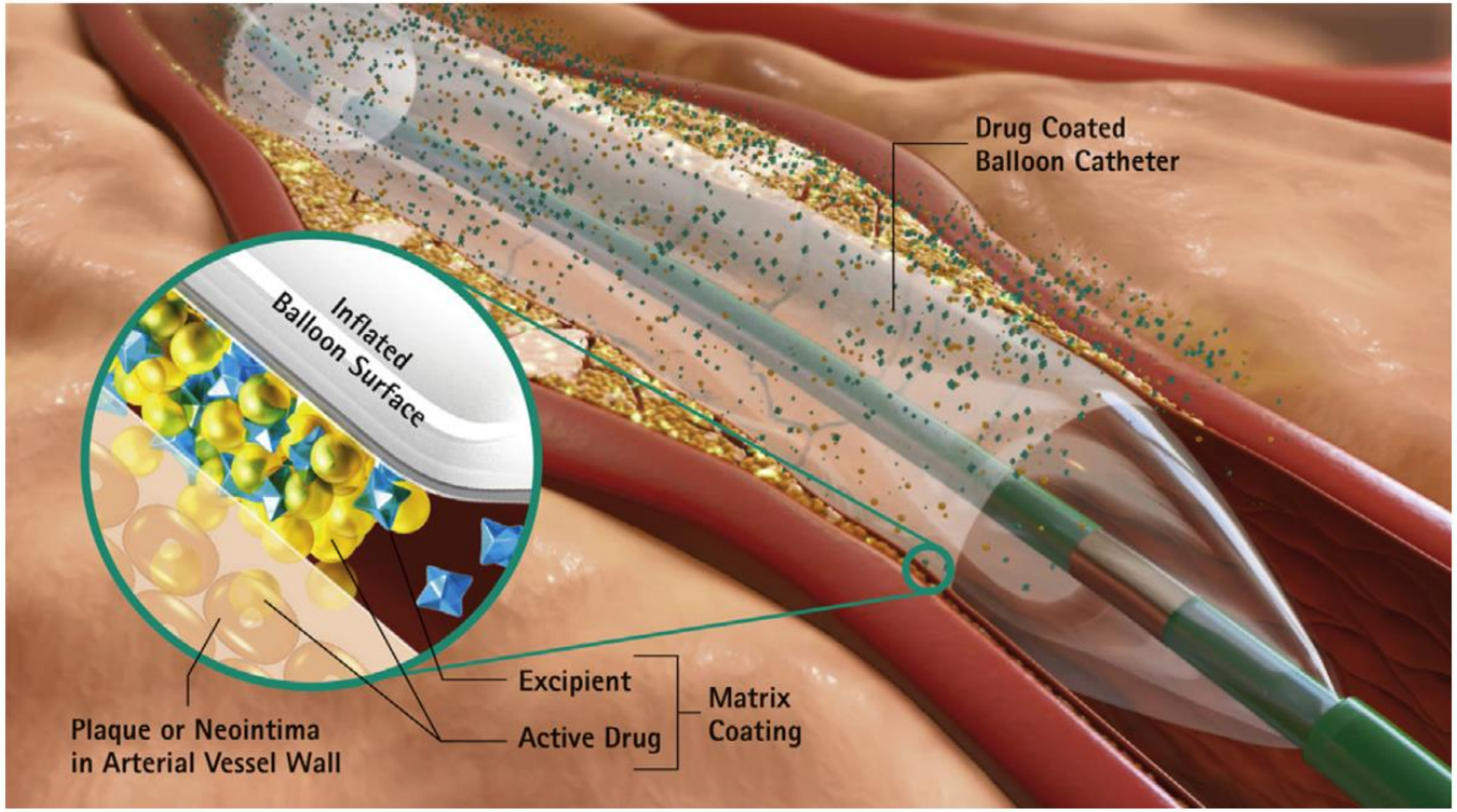
- **Background**
- **REC-CAGEFREE Trial**
- **AGENT IDE Trial**

Background

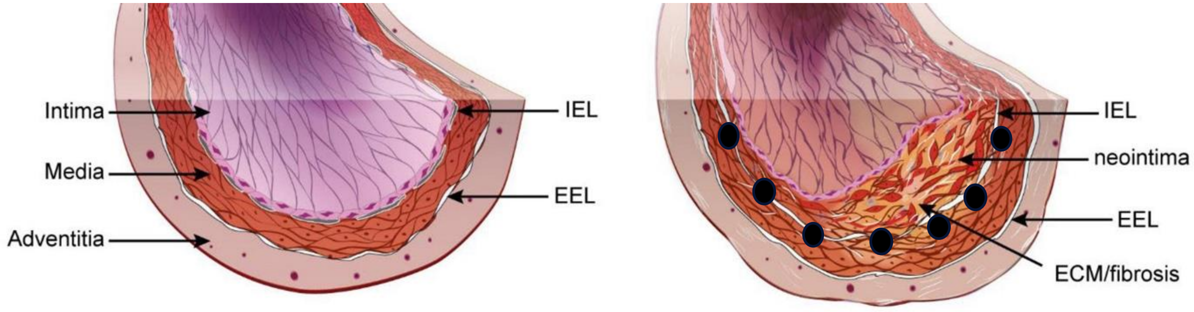
What is a drug-coated balloon (DCB)?

- Semi-compliant balloon coated with a pharmacologically active, lipophilic matrix layer including the **antiproliferative drug** and an **excipient** (inactive substance that contributes to drug transfer and tissue retention)

- Concept first introduced in 2000 based on positive results in animal studies demonstrating reduction in neointimal hyperplasia, with first positive human studies for ISR occurring in 2006



Jeger et al. Circulation: Cardiovascular Interventions 2020



healthy vessel

intimal hyperplasia

Antiproliferative Drug

- **Paclitaxel**
  - most widely used in DCB technologies because of its lipophilicity and efficient cellular uptake
  - Cytotoxic
  - decreases smooth muscle proliferation, migration, and formation of ECM (i.e., neointima formation) by irreversibly binding to the β subunit of tubulin, halting microtubule disassembly and inhibiting cell-cycle progression
  - Drug retention ~90 days

- **Sirolimus**
  - most commonly used limus-based analogue in DES; utilized in some DCB technologies
  - cytostatic
  - less lipophilic, poorer transfer rate
  - reversible inhibition of vascular smooth muscle cell cycle progression

Rationale for use of DCB

- While they are the gold standard in the modern era of revascularization, a universal DES implantation strategy holds limitations:
    1. stents inhibit positive vascular remodeling and pulsatile function
    2. 2% yearly accrual rate in adverse events; higher for long stents or those with diabetes
    3. stent and device delivery across a stent can be very complex in tortuous anatomies
    4. consequences of suboptimal stent implantation can be catastrophic
    5. in-stent restenosis (ISR) can be very difficult to treat

- DCB offer a "cage-free" option and may accelerate arterial healing, preserve normal vessel anatomy and function, and offer the potential for positive vessel remodeling

Figure 4. Potential Applications of Drug-Coated Balloon (DCB) Angioplasty

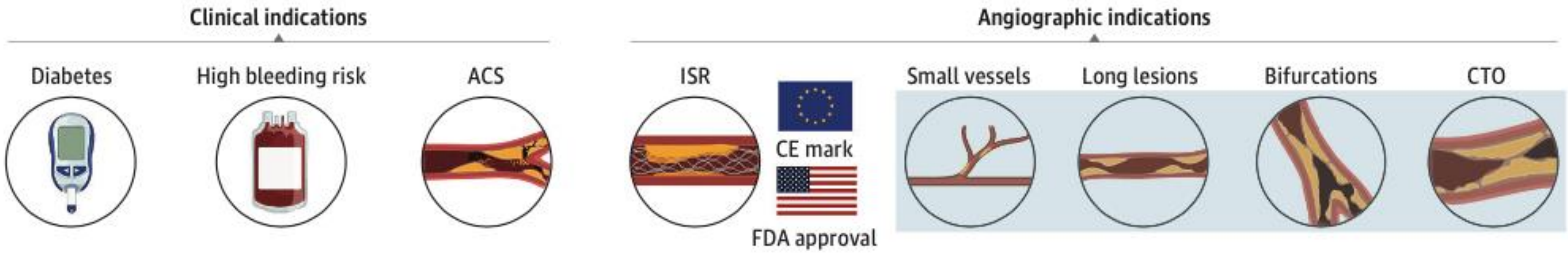


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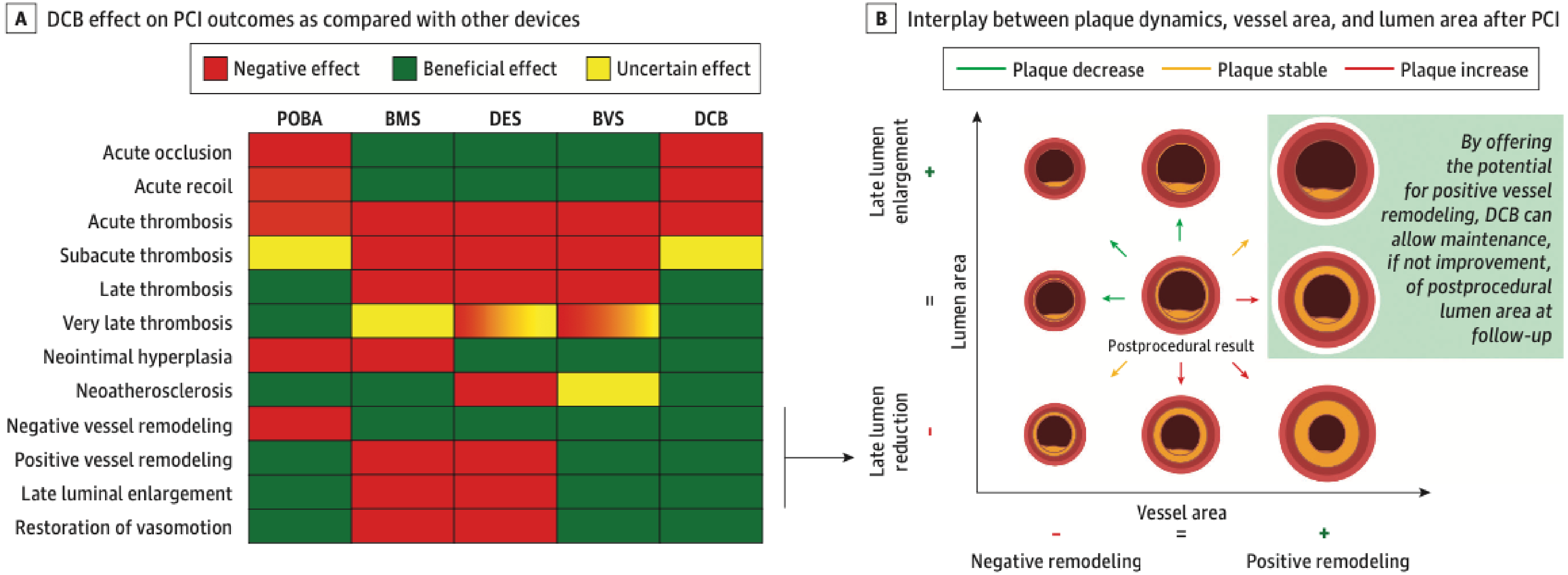
Figure 1. Drug-Coated Balloon (DCB) Effect as Compared With Other Devices and Vessel Remodeling Scenarios After DCB



DCB effect on PCI outcomes as compared with other devices. Interplay between plaque dynamics, vessel area, and lumen area after percutaneous coronary intervention (PCI). A comparison of potential PCI outcomes according to the type of device adopted is presented on the left. The impact of different vessel remodeling scenarios on residual lumen area is depicted on the right. Relative changes in vessel area (vessel remodeling) and plaque burden determine lumen area. The possibility to avoid late lumen reduction or gain late lumen enlargement with DCB derives from the potential positive vessel remodeling in absence of permanent stent implantation. BMS indicates bare metal stent; BVS, bioresorbable vascular scaffold; DES, drug-eluting stent; POBA, plain balloon angioplasty.

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Potential use scenarios for DCB

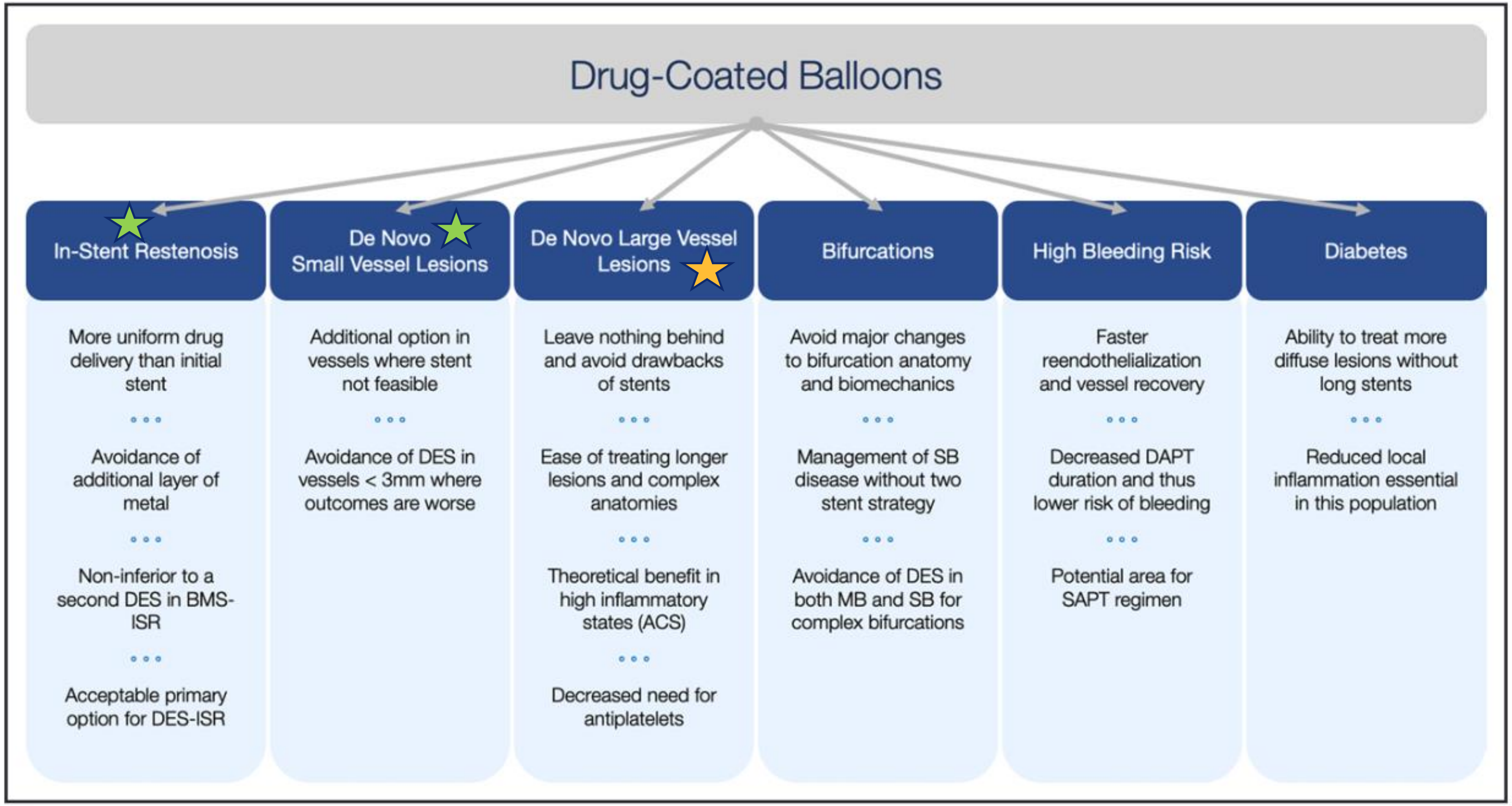

Figure. Features of drug-coated balloons and potential use scenarios.
ACS indicates acute coronary syndrome; BMS, bare metal stent; DAPT, dual antiplatelet therapy; DES, drug-eluting stent; ISR, in-stent restenosis; MB, main branch; SAPT, single antiplatelet therapy; and SB, side branch.

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Current guidelines

- **No US guidelines**
  - 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes
  - 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients With Chronic Coronary Disease
  - 2021 ACC/AHA/SCAI Guideline for Coronary Artery Revascularization

- **ESC**
  - 2014 ESC/EACTS Guidelines on myocardial revascularization: The Task Force on Myocardial Revascularization of the European Society of Cardiology (ESC) and the European Association for Cardio-Thoracic Surgery (EACTS)

Restenosis			
Repeat PCI is recommended, if technically feasible.	I	C	
DES are recommended for the treatment of in-stent restenosis (within BMS or DES).	I	A	501,502,508 511,524
Drug-coated balloons are recommended for the treatment of in-stent restenosis (within BMS or DES).	I	A	507– 511,524
IVUS and/or OCT should be considered to detect stent-related mechanical problems.	IIa	C	

Currently Available DCBs

FDA Approval 2/29/24

Table 1. Currently Available DCBs for the Treatment of Coronary Artery Disease (Table view)

Device	Drug	Dose, $\mu g/mm^2$	Excipient	Availability
Agent Boston Scientific	Paclitaxel	2.0	ATBC	Europe, Asia, United States, and Latin America
BioStream Biosensors International	Paclitaxel	3.0	Shellac	Europe
Danubio Minivasys	Paclitaxel	2.5	BTHC	Europe
Dior Eurocor	Paclitaxel	3.0	Shellac	Europe, Asia
Elutax SV Aachen Resonance	Paclitaxel	2.2	Dextran coating without excipient	Europe
Essential Pro iVascular	Paclitaxel	3.0	Organic ester	Europe, Asia
Magic Touch Concept Medical	Siroliumus	1.3	Phospholipid	Europe, Asia
Pantera Lux Biotronik	Paclitaxel	3.0	BTHC	Europe, Asia
Prevail Medtronic	Paclitaxel	3.5	Urea	Europe
Protégé Wellinq	Paclitaxel	3.0	Hydrophilic coating without excipient	Europe
RESTORE Cardionovum	Paclitaxel	3.0	Shellac	Europe, Asia
Selution Med Alliance	Siroliumus	1.0	PLGA	Europe, Asia, and Latin America
SeQuent Please Neo B.Braun	Paclitaxel	3.0	Iopromide	Europe, Asia, and Latin America
SeQuent SCB B.Braun	Siroliumus	4.0	BHT	Europe, Asia, and Latin America
Virtue SAB Orchestra BioMed	Siroliumus	…	Nanoparticles without excipient	Europe, Asia

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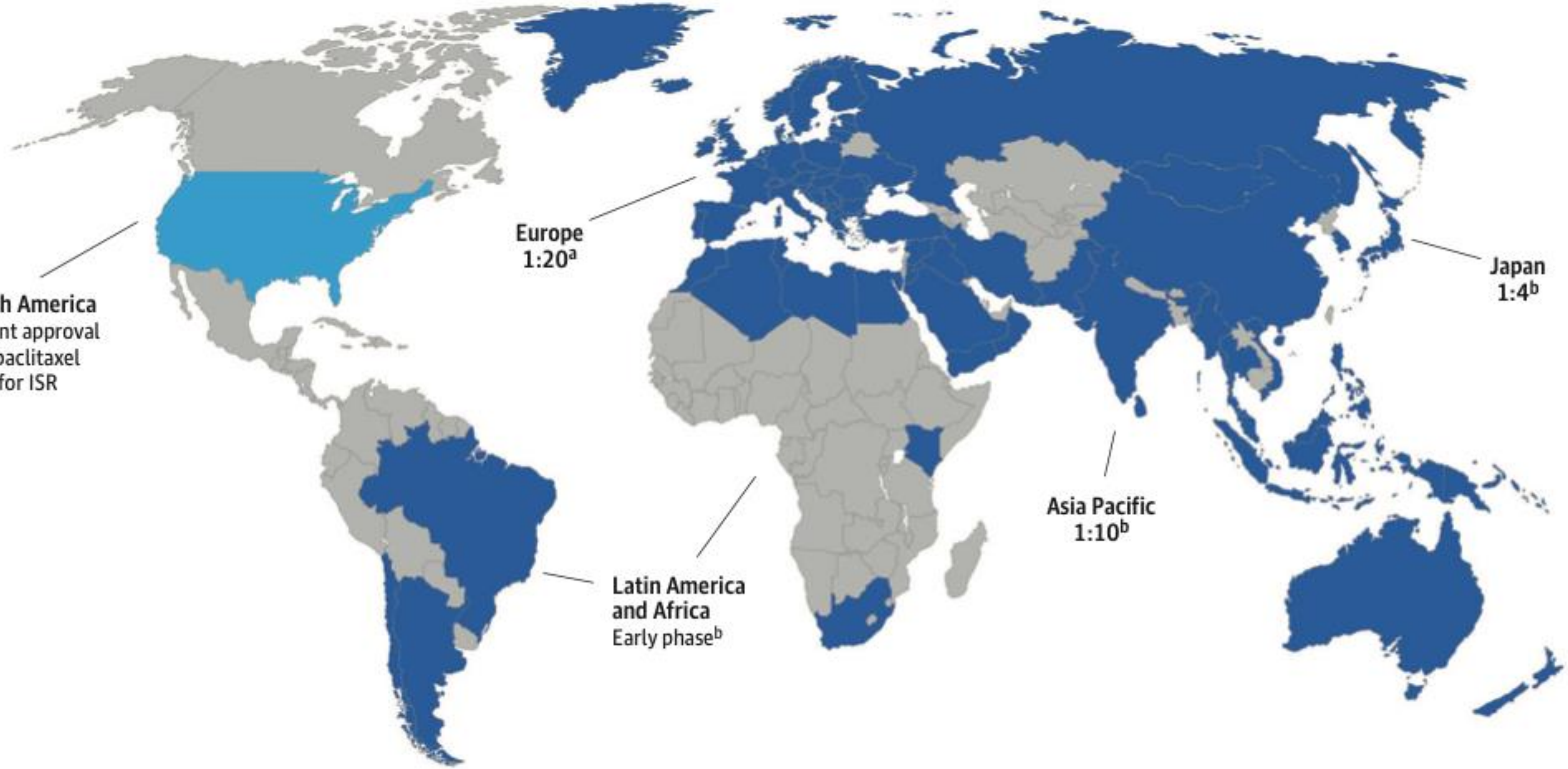
Contemporary use of DCB

Figure 2. Paclitaxel-Coated Balloon Usage Across the World



World distribution of a paclitaxel drug-coated balloon (DCB):drug-eluting stent (DES) ratio for percutaneous coronary intervention (PCI). The map outlines the penetration of a paclitaxel DCB (SeQuent Please [B. Braun]) usage across the globe. Usage of DCB vs DES for PCI is expressed as DCB:DES ratio. Figure readapted from B. Braun Melsungen AG. EUCOMED indicates European database on medical devices; ISR, in-stent restenosis.

[a]EUCOMED-Data Europe, 2022.

[b]EUCOMED-Data, 2022 & Internal Data B. Braun Melsungen AG.

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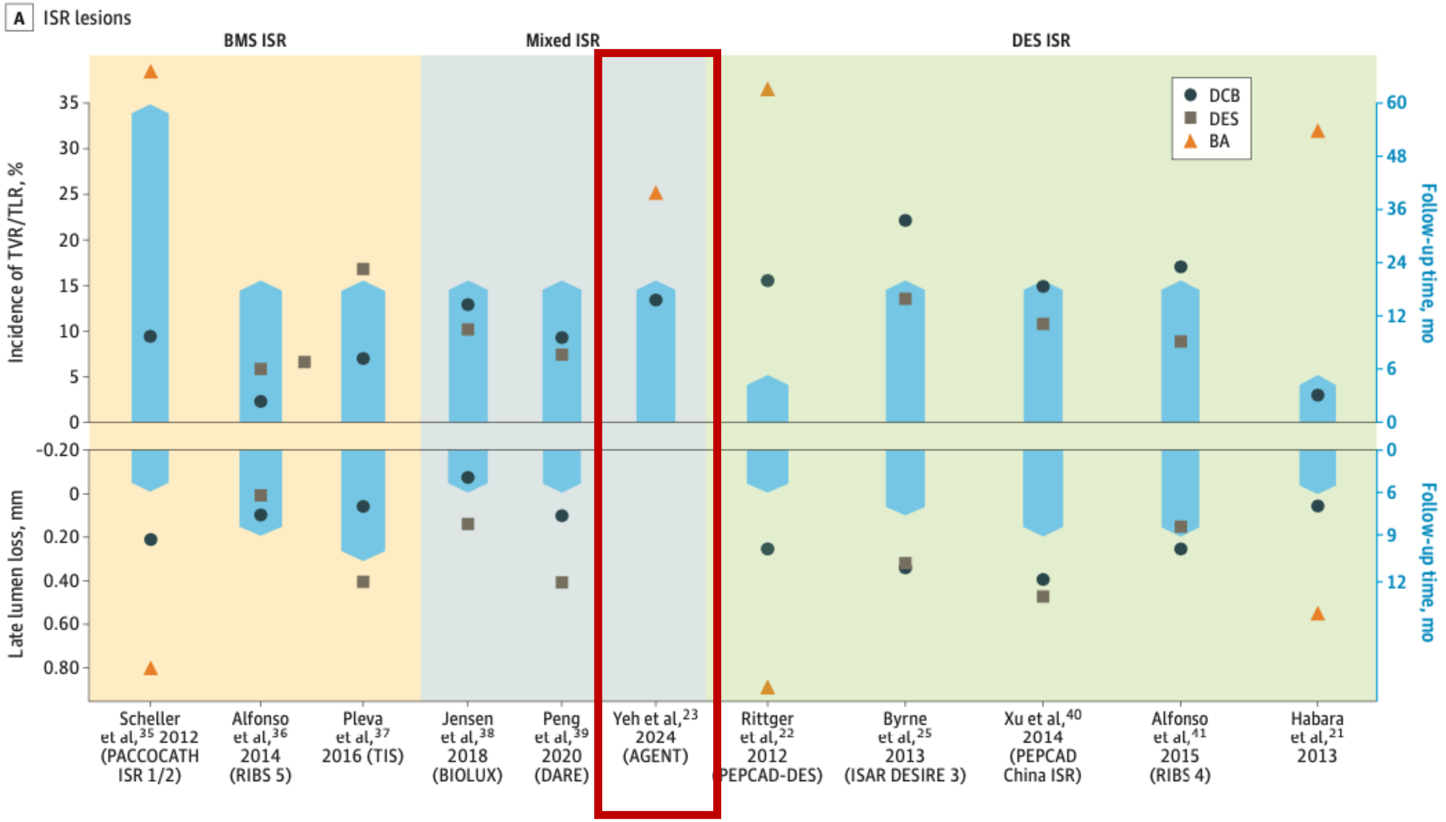
Indications for use: ISR

- Requires consideration of:
  - Underlying stent characteristics, i.e., BMS vs DES
  - Alternative PCI options, i.e., DCB vs POBA vs DES



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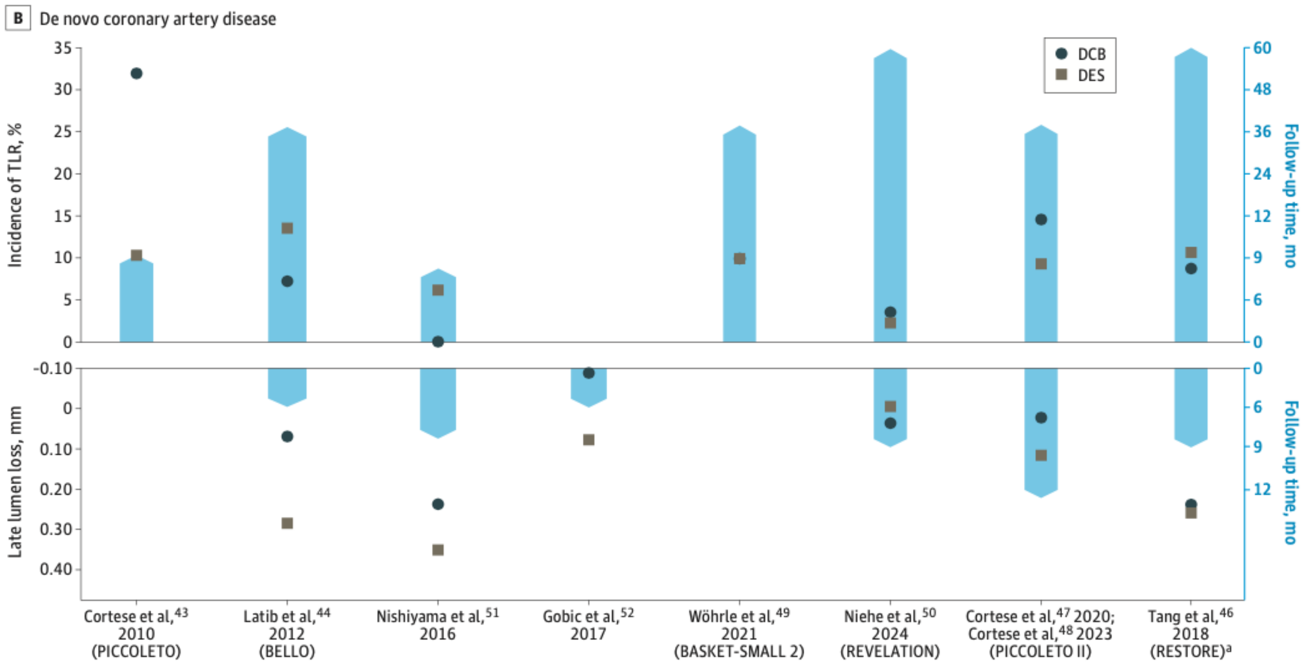
Indications for use: de novo lesions

- Most robustly studied in scenarios of CAD involving small vessels (<2.75 or 3 mm) and long lesions



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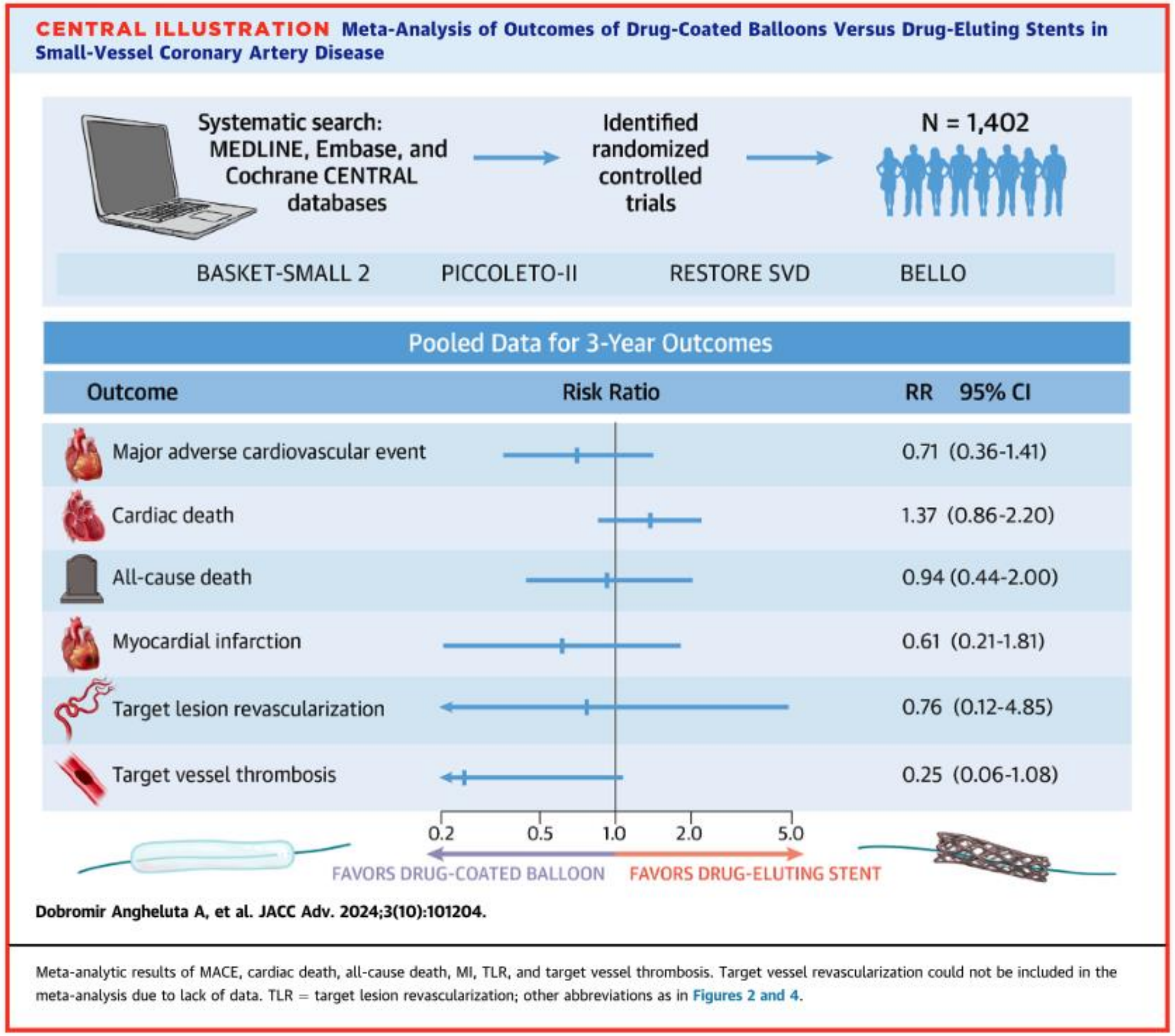

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TABLE 1 Summary of Clinical Trials of DCB on Treatment of De Novo Coronary Lesions

Trial Name or First Author Design	Treatment	n	Angiographic Follow-up	Clinical Follow-up
Small vessel disease				
BASKET-SMALL 2[15]/RCT	DCB (Sequent Please) vs DES (Xience or Taxus Element)	382 vs 376	-	MACE: 15% (DCB) vs 15% (DES), HR: 0.99; $P = 0.95$ at 3 y
PICCOLETO II[57]/RCT	DCB (Elutax SV) vs DES-E	118 vs 114	LLL: 0.04 mm (DCB) vs 0.17 mm (DES); $P = 0.001$ for noninferiority; $P = 0.03$ for superiority at 6 mo	MACE: 6% (DCB) vs 8% (DES); $P = 0.55$ at 12 mo
Large vessel disease				
Shin et al[28]/registry-observational	DCB (Sequent Please) vs BMS (Vision) or DES (Resolute Integrity or Xience Prime)	45 vs 22	LLL: 0.05 ± 0.27 mm (DCB) vs 0.40 ± 0.54 mm (Stent); $P = 0.022$; FFR: 0.85 ± 0.08 mm (DCB) vs 0.85 ± 0.05 mm (Stent); $P = 0.973$ at 9 mo	-
Her et al[9]/registry-observational	DCB (nonsmall) (Sequent Please) vs DCB (small) (Sequent Please)	100 vs 127	LLL: 0.03 ± 0.22 mm (nonsmall) vs 0.06 ± 0.25 mm (small); $P = 0.384$ at 6 mo	TVF: 7% (nonsmall) vs 8% (small); $P = 0.596$ at 6 mo
Gitto et al[64]/registry-PS matched	DCB-based (Magic Touch or Selution or IN.PACT Falcon or RESTORE) vs DES (contemporary)	147 vs 701		TLF: 4.1% (DCB-based) vs 9.8% (DES); $P = 0.148$ at 2 y
Lin et al[65]/meta-analysis	DCB vs DES	152 vs 169	LLL: SMD: −0.07; $P = 0.548$ at 6-9 mo	TLR: RR: 1.17; $P = 0.746$ at 6-9 mo
REC-CAGEFREE I[66]/RCT	DCB (Swide) vs DES-S	1,133 vs 1,139		DoCE: 6.4% (DCB) vs 3.4% (DES); $P = 0.0008$ at 2 y
Bifurcation disease				
PEPCAD-BIF[68]/RCT	DCB (Sequent Please) vs POBA in SB	32 vs 32	Restenosis: 6% (DCB) vs 26% (POBA); $P = 0.045$; LLL: 0.13 mm (DCB) vs 0.51 mm (POBA); $P = 0.013$ at 9 mo	TLR: n = 1 (DCB) vs n = 3 (POBA)
DCB-BIF[70]/RCT	DCB (Paclitaxel-coated balloon) vs POBA (NCB) in SB	391 vs 393		MACE: 7.2% (DCB) vs 12.5% (POBA); $P = 0.013$ at 1 y
Her et al/registry-observational[71]	DCB (only MV) (Sequent Please)	16 MBs/32 SBs	SB os lumen area: 0.92 ± 0.68 mm^2 (pre) vs 1.03 ± 0.77 mm^2 (post) vs 1.42 ± 1.18 mm^2 (at 9 mo)	-
Pan et al[72]/registry-PS matched	DCB (Sequent Please) + DES vs DES-only (new-generation)	199 vs 398	LLL: 0.13 ± 0.42 mm (DCB) vs 0.42 ± 0.62 mm (DES); $P < 0.001$ at 1 y	TLF: 7.56% (DCB) vs 14.36% (DES), log-rank $P = 0.024$ at 2 y; TLR: 2.91% (DCB) vs 9.42% (DES); $P = 0.007$
Chronic total occlusion lesion				
PEPCAD-CTO[77]/registry-variable matching	DCB (Sequent Please) + BMS (Coroflex Blue) vs DES-P	48 vs 48	LLL: 0.64 ± 0.69 mm (DCB) vs 0.43 ± 0.64 mm (DES); $P = 0.14$; Restenosis: 28% (DCB) vs 21% (DES); $P = 0.44$ at 6 mo	MACE: 15% (DCB) vs 19% (DES); $P = 0.58$ at 12 mo
Shin et al[79]/registry-PS matched	DCB-based (Sequent Please) vs DES-only (2nd generation)	200 vs 661		MACE: 3.1% (DCB) vs 13.2% (DES); $P = 0.001$ at 2 y
Diffuse long and multivessel disease				
Costopoulos et al[80]/registry-variable matching	DCB (IN.PACT Falcon + Pantera Lux) ± DES vs DES (2nd generation)	93 vs 93	-	MACE: 21% (DCB) vs 23% (DES); $P = 0.74$; TVR: 15% (DCB) vs 12% (DES); $P = 0.44$; TLR: 10% (DCB) vs 9% (DES); $P = 0.84$ at 2 y
Yang et al[81]/registry-observational	DCB (Sequent Please) ± DES (new-generation) vs DES (new-generation)	360 vs 831	LLL: 0.06 ± 0.61 mm (DCB) vs 0.41 ± 0.64 mm (DES); $P < 0.001$; MLD: 1.74 ± 0.59 mm (DCB) vs 1.94 ± 0.65 mm (DES); $P = 0.002$ at 9-12 mo	TLR: 7% (DCB) vs 8% (DES), log-rank $P = 0.636$; MACE: 11% (DCB) vs 14% (DES), log-rank $P = 0.324$; Thrombosis: 0 (DCB) vs 1% (DCB), log-rank $P = 0.193$ at 3 y
Shin et al[44]/registry-PS matched	DCB-based (Sequent Please) vs DES (2nd generation)	254 vs 254	-	MACE: 4% (DCB) vs 11% (DES); $P = 0.002$ at 2 y

Her et al. JAMA Asia 2025

REC-CAGEFREE I

THE LANCET

ARTICLES · Volume 404, Issue 10457, P1040-1050, September 14, 2024

Drug-coated balloon angioplasty with rescue stenting versus intended stenting for the treatment of patients with de novo coronary artery lesions (REC-CAGEFREE I): an open-label, randomised, non-inferiority trial

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Study Design / Methods

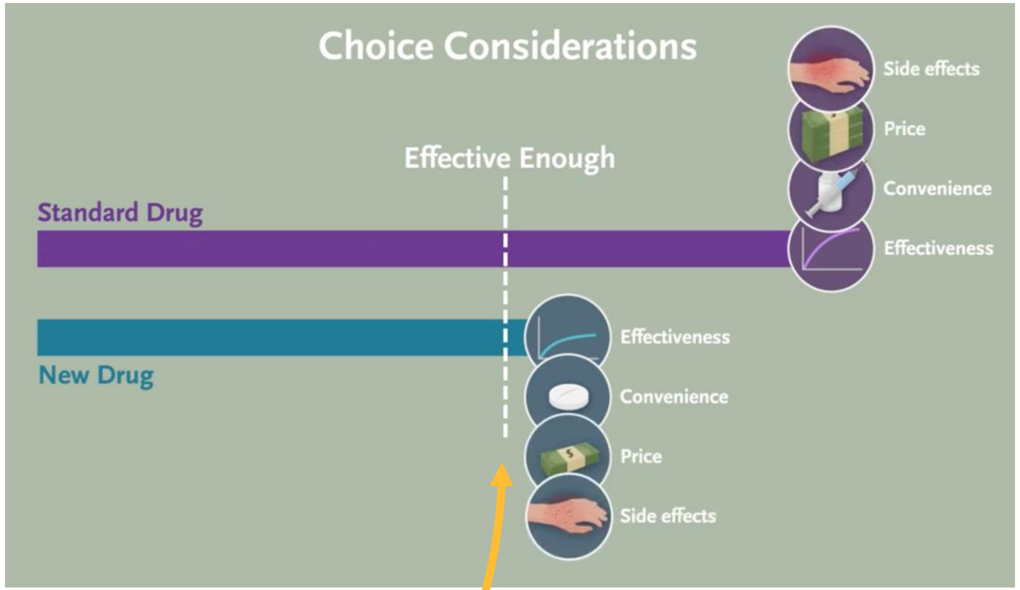
- open-label, randomized, intention-to-treat, **non-inferiority** trial
  - “A **non-inferiority margin of 2.68%** was chosen based on clinically acceptable relevance according to the margins in previous pivotal trials comparing DCB with DES or one DES with another.”
  - Patients and enrolling investigators were not masked to treatment allocation
  - Members of the independent clinical event committee (CEC) who adjudicated endpoints and the statisticians were masked to treatment allocation

- conducted at 43 sites across China

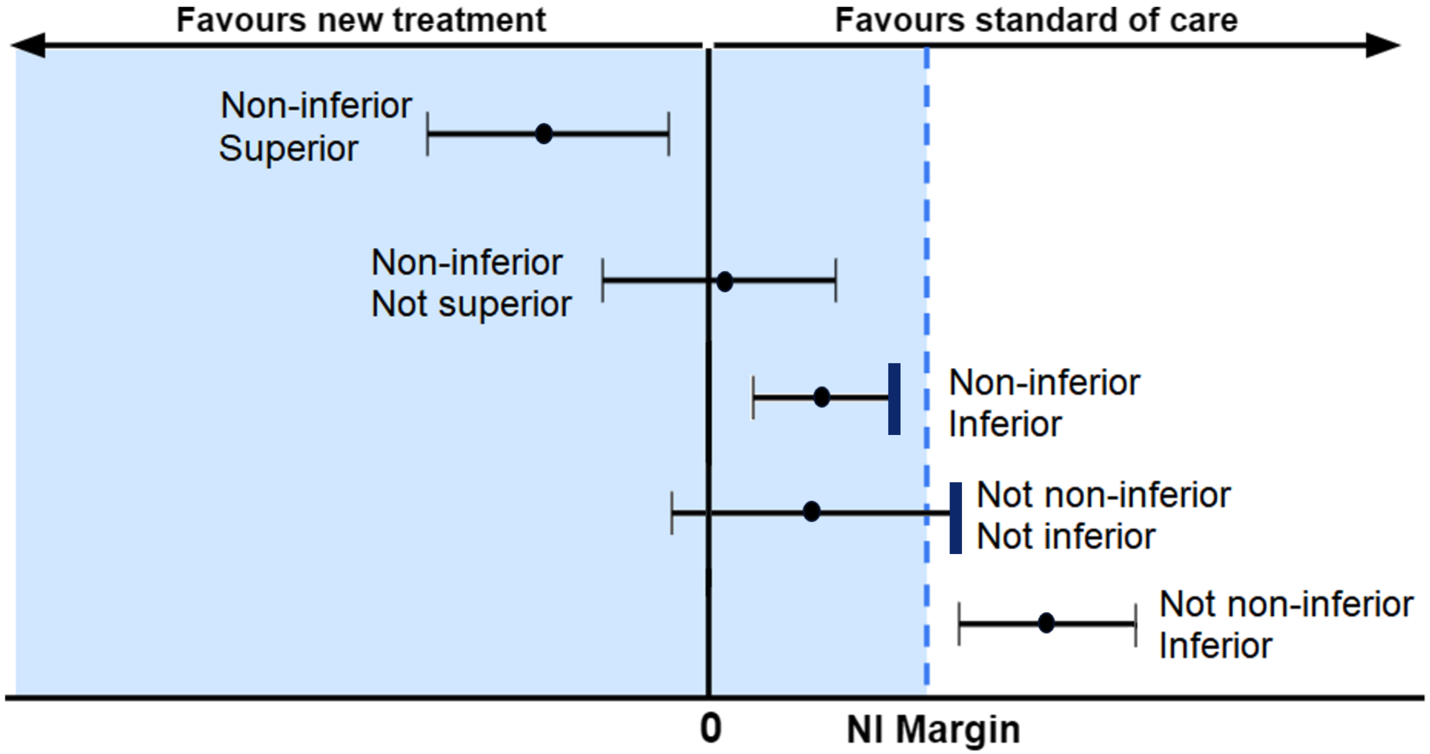
- enrollment between Feb 5, 2021, and May 1, 2022

- Paclitaxel-coated Swide DCB (Shenqi Medical, Shanghai, China)

Non-inferiority Trial Design & Outcomes



Non-inferiority
margin



2.68%

Study Design / Methods

- **Inclusion Criteria:**
  - Patients aged 18 years or older requiring PCI with “acute or chronic coronary syndrome”
  - de novo, <u>non-complex</u> target lesions, regardless of treated vessel diameter
    - 1-2 target lesions or vessels
    - 1-2 planned DES or DCB devices, or planned total DES or DCB length <60 mm
    - bifurcations not requiring treatment in both main and side branch
    - lesions not requiring atherectomy

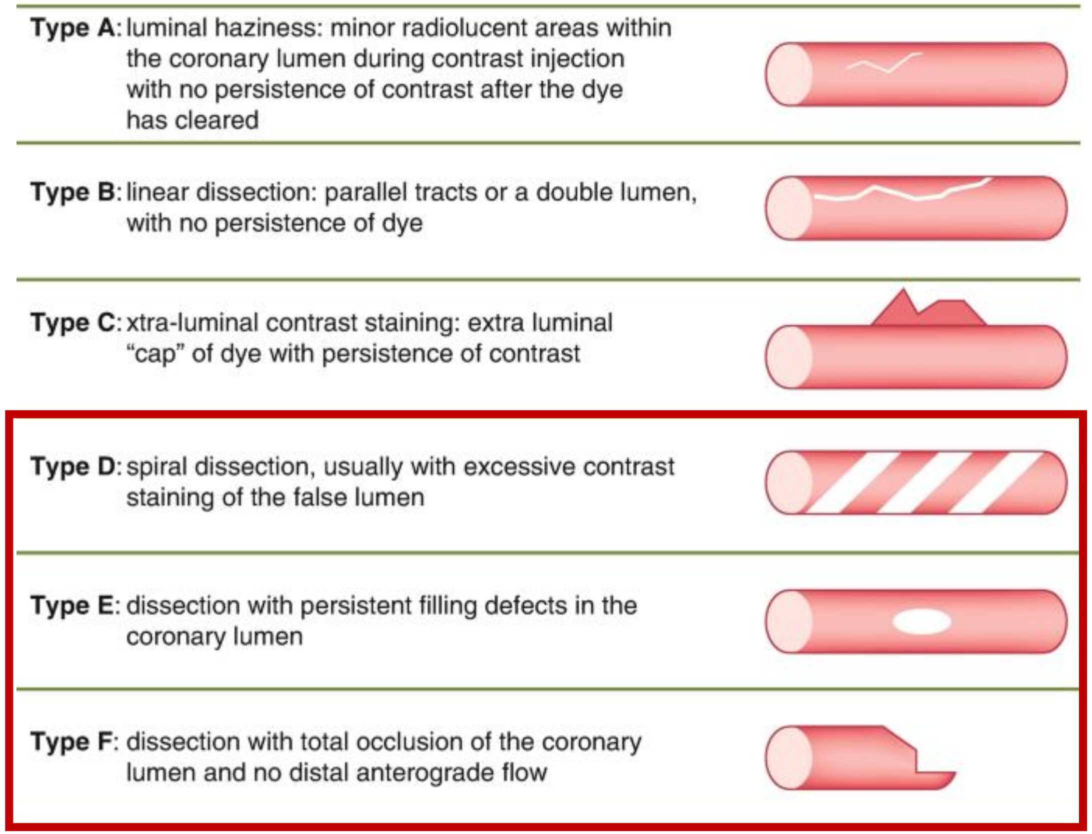
- **Key exclusion criteria:**
  - cardiogenic shock
  - ISR requiring revascularization
  - left main, venous/arterial graft, or CTO lesions (*considered complex target lesions*)

Study Design / Methods

- Randomization (1:1) was conducted after successful and satisfactory pre-dilatation of the target lesion, defined as:
  - no type D, E, or F dissection
  - no decreased blood flow (TIMI flow 3)
  - no residual stenosis of 30% or more by visual assessment
  - no serious complications requiring termination of the PCI

- Patients with unsuccessful pre-dilatation were not randomly assigned to treatment but were included in a separate nested registry.

National Heart, Lung, and Blood Institute Dissection Classification System



Study Design / Methods

- **DCB group**
  - procedural techniques followed the recommendations of the German Consensus Group on DCB interventions and the Third Report of the International DCB Consensus Group
  - After DCB angioplasty, rescue stenting (which used the same stent type as for the DES group) was mandated for patients with:
    - type D, E, or F dissections
    - TIMI flow less than 3
    - visual residual stenosis of more than 30%

- **DES group**
  - procedural techniques followed routine local clinical practice and established guidelines
  - patients received the Firebird 2 DES (MicroPort, Shanghai, China); a sirolimus-eluting stent

Study Design / Methods

- All participants received DAPT for at least 1-month post-PCI
  - aspirin (100 mg) + clopidogrel (75 mg) / aspirin + ticagrelor (90 mg twice daily)
  - decision re: full duration of DAPT was at the discretion of the implanting physician

- Patients then transitioned to monotherapy with aspirin (100 mg) or clopidogrel (75 mg)

- Physicians had discretion over other medical treatments depending on each patient's specific condition, but guideline-directed medical therapy was strongly recommended.

Study Design / Methods

- Follow-up visits occurred at 1 (± 14 days), 3, 6, 12, 18, and 24 (± 30 days) months after randomization

- Visits were preferably conducted on site; however, the scheduled visit could be replaced by a telephone call, except for the 1-month, 1-year, and 2-year visits

- The total length of follow-up prespecified in the protocol is 10 years, with annual follow up after the initial 2 years

- A mobile application operating on the WeChat platform was developed to assess patients' health status (e.g., to report symptoms of potential adverse events), with participants contacted through this application monthly in the first 2 years of follow-up and then every 3 months

Study Design / Methods

- **Primary end point:** device-oriented composite endpoint (DoCE)
  - composite of cardiovascular death, target vessel myocardial infarction (TV-MI), and target lesion revascularization (TLR), assessed at <u>24 months</u>
  - TLR was considered clinically and physiologically indicated if associated with wire-based or angiography-derived FFR <0.81, or a diameter stenosis of >70% by quantitative coronary angiography

- **Secondary end points:**
  - rates of DoCE at 1 month and 12 months
  - rates of the individual components of the DoCE
  - a patient-oriented composite endpoint (all-cause death, any stroke, any myocardial infarction, and any revascularization)
  - target vessel failure (defined as vessel-related cardiovascular death, TV-MI, TLR)
  - Bleed events, using BARC definitions
  - definite or probable device or vessel thrombosis
  - device / procedure success

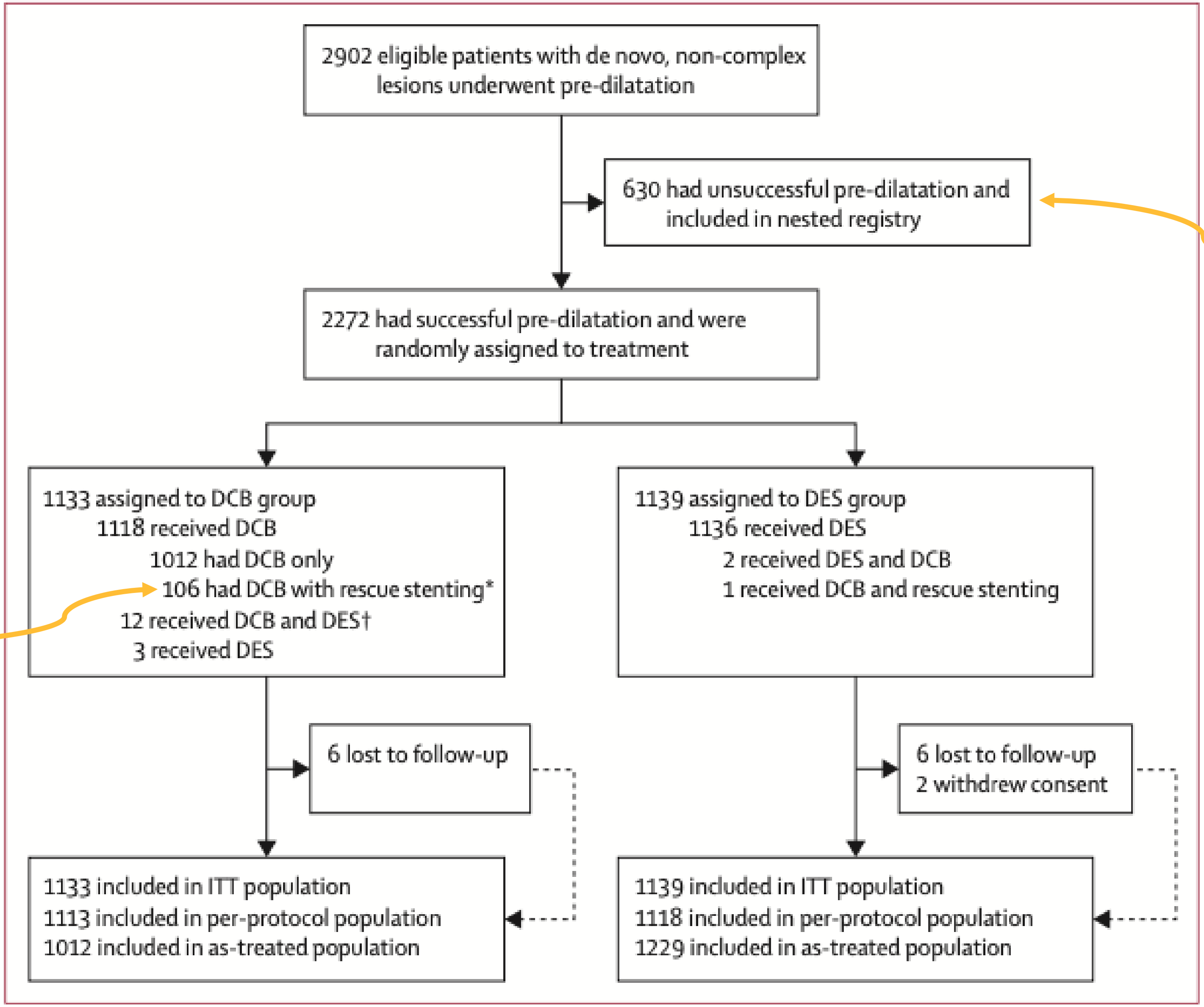
Study Design / Methods

- Sample size and power calculations were based on a non-inferiority assumption for the primary outcome, with the non-inferiority margin set at 2.68%
  - Non-inferiority would be concluded if the upper limit of the one-sided 95% CI in the treatment difference of the primary endpoint was <2.68%

- According to data from previous trials, authors assumed that 6.7% of patients in the intended DES group would reach the primary endpoint at 2 years.

- Considering an anticipated patient attrition rate of 5%, 2270 patients were required for the study to have 80% power to show non-inferiority with a 5% one-sided α error rate.

Sample Population



21.7%

9.5%

Sample Baseline Characteristics

	DCB group (n=1133)	DES group (n=1139)
Demographics		
Age, years	62 (55–69)	62 (54–69)
Sex		
Male	769 (67·9%)	805 (70·7%)
Female	364 (32·1%)	334 (29·3%)
BMI, kg/m²	24·6 (22·5–26·8)	24·4 (22·5–27·0)
Smoking		
Former	123 (10·9%)	125 (11·0%)
Current	360 (31·8%)	404 (35·5%)
Never	650 (57·4%)	610 (53·6%)
Comorbid conditions		
Arterial hypertension	662/1133 (58·4%)	704/1139 (61·8%)
Diabetes	282/1133 (24·9%)	338/1139 (29·7%)
Insulin-treated diabetes	67/282 (23·8%)	73/338 (21·6%)
Hyperlipidaemia	903/1133 (79·7%)	908/1139 (79·7%)
Left ventricular ejection fraction ≤40% or previous episode of heart failure	59/1133 (5·2%)	55/1139 (4·8%)
Left ventricular ejection fraction, %*	60 (56–65)	60 (56–65)
Previous stroke	94/1133 (8·3%)	95/1139 (8·3%)
Previous myocardial infarction	81/1133 (7·1%)	105/1139 (9·2%)
Previous PCI	137/1133 (12·1%)	142/1139 (12·5%)
COPD	62/1130 (5·5%)	60/1134 (5·3%)
Chronic kidney disease†	95/1133 (8·4%)	108/1139 (9·5%)
Peripheral vascular disease	41/1133 (3·6%)	59/1139 (5·2%)

	DCB group (n=1133)	DES group (n=1139)
Clinical presentation		
ST-elevation myocardial infarction	181 (16·0%)	185 (16·2%)
Non-ST-elevation myocardial infarction	214 (18·9%)	201 (17·6%)
Unstable angina	235 (20·7%)	241 (21·2%)
Chronic coronary syndrome	503 (44·4%)	512 (45·0%)
PCI characteristics		
Number of lesions treated per patient		
Mean (SD)	1·1 (0·3)	1·1 (0·3)
Median (IQR)	1 (1–1)	1 (1–1)
Number of DCB or DES per patient		
Mean (SD)	1·4 (0·7)	1·3 (0·5)
Median (IQR)	1 (1–2)	1 (1–2)
Length of DCB or DES per patient, mm	30 (20–40)	29 (18–33)
IVUS or OCT guidance for PCI	108 (9·5%)	127 (11·2%)
Contrast material used, mL	120 (100–160)	130 (100–170)
Procedure time, min	34 (23–50)	36 (25–50)
SYNTAX score		
Mean (SD)	7·3 (4·7)	7·4 (4·9)
Median (IQR)	7 (4–9)	7 (4–9)
Residue SYNTAX score		
Mean (SD)	1·4 (3·0)	1·4 (2·8)
Median (IQR)	0 (0–1)	0 (0–2)

(Table 1 continues on next page)

	DCB group (n=1133)	DES group (n=1139)
(Continued from previous page)		
Lesion characteristics		
Total number	1266	1263
Lesion location		
Left anterior descending artery	656/1266 (51·8%)	626/1263 (49·6%)
Proximal left anterior descending artery	332/1266 (26·2%)	356/1263 (28·2%)
Left circumflex artery	292/1266 (23·1%)	316/1263 (25·0%)
Right coronary artery	318/1266 (25·1%)	321/1263 (25·4%)
Pre-dilatation balloon types		
Semi-compliant balloon	1000/1266 (79·0%)	974/1263 (77·1%)
Non-compliant balloon	523/1266 (41·3%)	453/1263 (35·9%)
Cutting or scoring balloon	861/1266 (68·0%)	770/1263 (61·0%)
Small vessel (<3·0 mm)	638/1266 (50·4%)	587/1263 (46·5%)
Bifurcation involved	410/1261 (32·5%)	388/1248 (31·1%)
Thrombus aspiration	37/1266 (2·9%)	37/1263 (2·9%)
Diameter of DCB or DES per device, mm	3·00 (2·75–3·50)	3·00 (2·75–3·50)
Inflation duration of DCB or DES, s	60 (50–60)	10 (6–10)

Data are n (%), n/N (%), or median (IQR), except where otherwise stated. Percentages might not add to 100% due to rounding. COPD=chronic obstructive pulmonary disease. DCB=drug-coated balloon. DES=drug-eluting stent. eGFR=estimated glomerular filtration rate. IVUS=intravascular ultrasound. OCT=optical coherence tomography. PCI=percutaneous coronary intervention. SYNTAX=the SYNergy between percutaneous coronary intervention with TAXus and cardiac surgery trial. *Data on the left ventricular ejection fraction were available for 1092 patients in the DCB group and for 1102 in the DES group. †Chronic kidney disease was defined as kidney damage (pathological abnormalities or markers of damage, including abnormalities in blood or urine tests or imaging studies) or an eGFR (by the Modification of Diet in Renal Disease formula) of <60 mL/min per 1·73 m² of body-surface area for at least 3 months. eGFR was available for 1069 patients in the DCB group and for 1083 in the DES group.

Table 1: **Baseline demographic and clinical characteristics in the intention-to-treat population**

* All patients were of Chinese ethnicity

Statistical Analysis

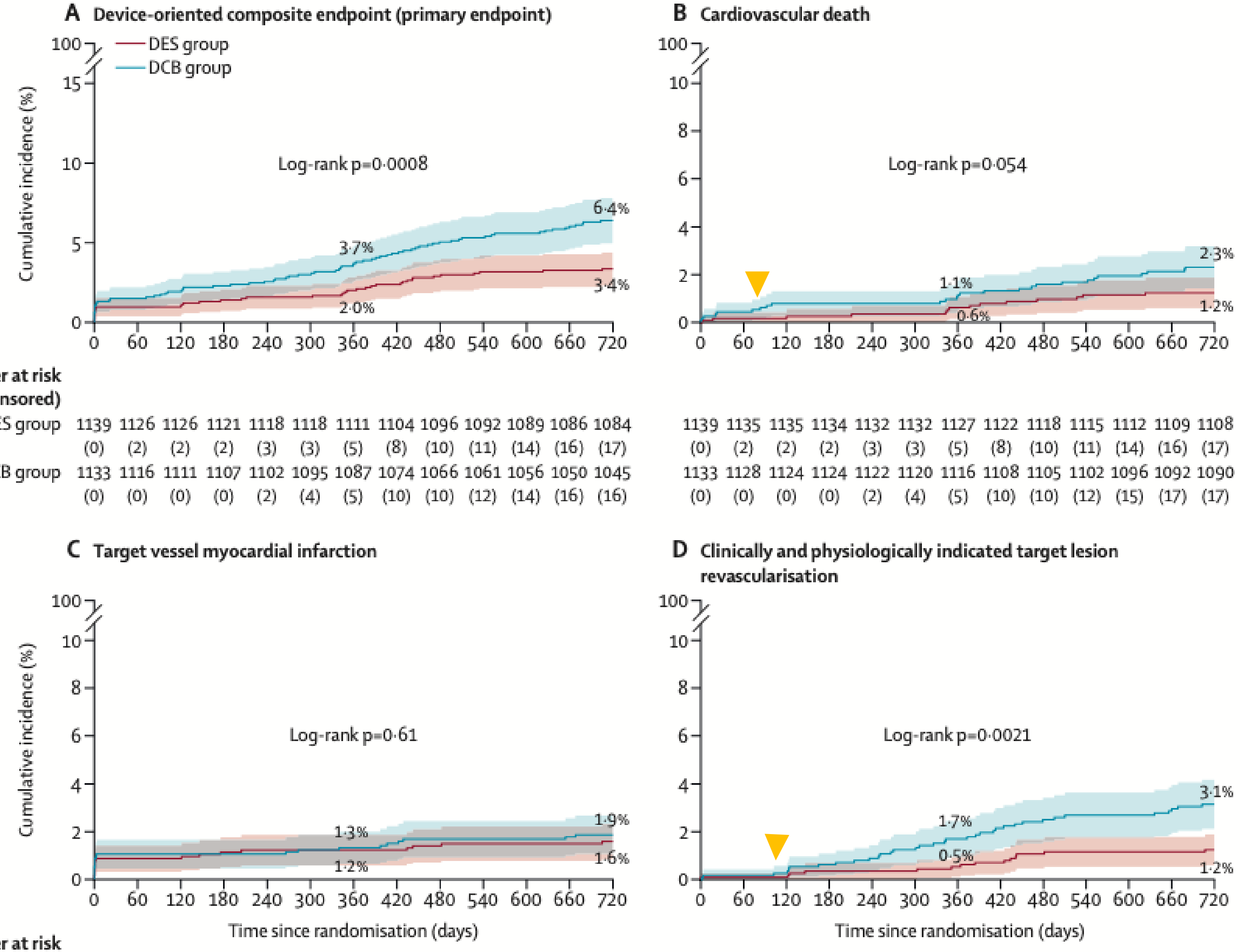
- Primary and secondary endpoints were analyzed in the **intention-to-treat (ITT) population**
  - estimated at 720 days by the **Kaplan–Meier method**, with standard error of difference calculated using Greenwood’s method and p value calculated using an approximate Z test

- All reported secondary endpoints were prespecified, but the two-sided 95% CIs were not adjusted for multiple comparisons, and so these intervals should not be used to infer definitive treatment effects.

- The secondary endpoints involving hierarchical orders were analyzed via the win ratio method

- Key prespecified subgroup analyses were done with Cox proportional hazards regression models, presented as hazard ratios with 95% CIs with p values

Results



Number at risk (number censored)

A

	0	60	120	180	240	300	360	420	480	540	600	660	720
DES group	1139 (0)	1126 (2)	1126 (2)	1121 (2)	1118 (3)	1118 (3)	1111 (5)	1104 (8)	1096 (10)	1092 (11)	1089 (14)	1086 (16)	1084 (17)
DCB group	1133 (0)	1116 (0)	1111 (0)	1107 (0)	1102 (2)	1095 (4)	1087 (5)	1074 (10)	1066 (10)	1061 (12)	1056 (14)	1050 (16)	1045 (16)

B

	0	60	120	180	240	300	360	420	480	540	600	660	720
DES group	1139 (0)	1135 (2)	1135 (2)	1134 (2)	1132 (3)	1132 (3)	1127 (5)	1122 (8)	1118 (10)	1115 (11)	1112 (14)	1109 (16)	1108 (17)
DCB group	1133 (0)	1128 (0)	1124 (0)	1124 (0)	1122 (2)	1120 (4)	1116 (5)	1108 (10)	1105 (10)	1102 (12)	1096 (15)	1092 (17)	1090 (17)

C

	0	60	120	180	240	300	360	420	480	540	600	660	720
DES group	1139 (0)	1126 (3)	1126 (3)	1123 (3)	1120 (5)	1120 (5)	1115 (10)	1110 (15)	1104 (19)	1100 (22)	1097 (25)	1094 (28)	1092 (29)
DCB group	1133 (1)	1116 (5)	1112 (9)	1112 (9)	1110 (11)	1106 (13)	1101 (18)	1092 (24)	1087 (27)	1084 (30)	1078 (36)	1073 (40)	1070 (42)

D

	0	60	120	180	240	300	360	420	480	540	600	660	720
DES group	1139 (0)	1134 (4)	1134 (5)	1130 (5)	1128 (7)	1128 (7)	1121 (12)	1114 (17)	1106 (21)	1102 (24)	1099 (27)	1096 (30)	1094 (31)
DCB group	1133 (1)	1126 (5)	1121 (9)	1117 (9)	1112 (11)	1106 (13)	1097 (18)	1084 (25)	1077 (28)	1072 (31)	1067 (36)	1061 (40)	1056 (42)

Results

lower than the anticipated 6.7% rate

	DCB group (n=1133)	DES group (n=1139)	Difference (95% CI)	p value
Primary endpoint				
Device-oriented composite endpoint*	72 (6·4%)	38 (3·4%)	3·04% (1·27 to 4·81)	0·0008
Secondary endpoints				
Patient-oriented composite endpoint	134 (11·9%)	90 (7·9%)	3·93% (1·48 to 6·39)	0·0017
Net adverse clinical events	138 (12·2%)	105 (9·2%)	2·97% (0·42 to 5·51)	0·022
Target vessel failure	74 (6·6%)	41 (3·6%)	2·95% (1·14 to 4·76)	0·0014
Any death	37 (3·3%)	23 (2·0%)	1·25% (–0·07 to 2·57)	0·064
Cardiovascular death	26 (2·3%)	14 (1·2%)	1·07% (–0·02 to 2·16)	0·053
Any stroke	15 (1·3%)	10 (0·9%)	0·46% (–0·41 to 1·32)	0·30
Any myocardial infarction†	23 (2·1%)	23 (2·0%)	0·02% (–1·14 to 1·19)	0·97
Q wave	3 (0·3%)	5 (0·4%)	–0·18% (–0·66 to 0·31)	..
Non-Q wave	20 (1·8%)	18 (1·6%)	0·20% (–0·87 to 1·26)	..
Target-vessel myocardial infarction	21 (1·9%)	18 (1·6%)	0·28% (–0·79 to 1·36)	0·61
Periprocedural myocardial infarction	10 (0·9%)	9 (0·8%)	0·09% (–0·66 to 0·84)	..
Any revascularisation	78 (7·0%)	52 (4·6%)	2·39% (0·45 to 4·32)	0·016
Target lesion revascularisation	49 (4·4%)	15 (1·3%)	3·07% (1·69 to 4·45)	..
Clinically and physiologically indicated	35 (3·1%)	14 (1·2%)	1·90% (0·69 to 3·11)	0·0021
Target vessel revascularisation	54 (4·8%)	20 (1·8%)	3·07% (1·60 to 4·55)	..
Clinically and physiologically indicated	37 (3·3%)	18 (1·6%)	1·73% (0·44 to 3·01)	0·0084
Non-target vessel revascularisation	31 (2·8%)	37 (3·3%)	–0·50% (–1·92 to 0·92)	..
Definite or probable device or vessel thrombosis	4 (0·4%)	3 (0·3%)	0·09% (–0·37 to 0·55)	0·69
BARC type 3 or 5 bleeding	16 (1·4%)	27 (2·4%)	–0·96% (–2·09 to 0·17)	0·096

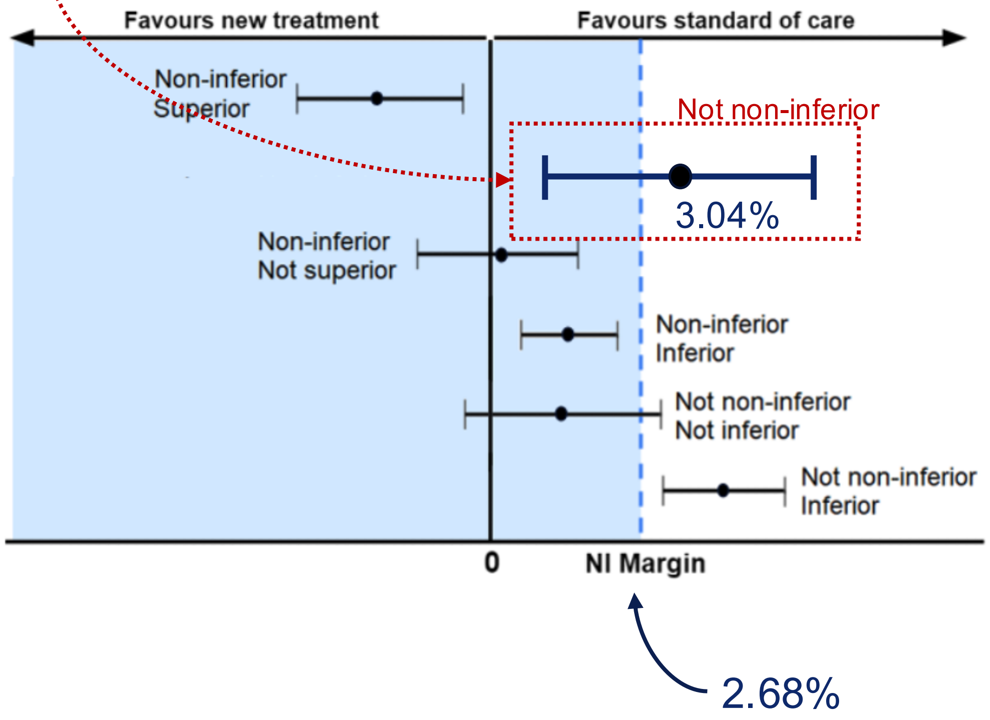
Data are n (cumulative event rate) or difference (95% CI), unless otherwise indicated. The listed cumulative event rates were estimated with the use of the Kaplan–Meier method, and so values might not be equivalent to those determined using the group denominator. Some components of prespecified endpoints are presented to aid understanding of the subclassification and features of the endpoints; since these were not prespecified endpoints, they are presented without p values. BARC=Bleeding Academic Research Consortium. DCB=drug-coated balloon. DES=drug-eluting stent. ITT=intention-to-treat. *For the between-group difference in the cumulative event rate of the primary outcome, the upper boundary of the one-sided 95% CI was 4·52 percentage points, and the p value for non-inferiority was 0·65. †Determined on the basis of the Society for Cardiovascular Angiography and Interventions 2013 definition within 48 h post-procedure or the fourth universal definition after 48 h post-procedure.

Table 2: **Primary and key secondary endpoints at 24 months after randomisation, ITT population**

***ITT**

Other primary endpoint analyses:

- Covariate-adjusted: 3.10% (1.34 - 4.86)
- Per-protocol: 3.08% (1.3 - 4.86)
- As-treated 1.96% (0.13 – 3.79)



Results

	n	Cumulative incidence (%) DCB group	Cumulative incidence (%) DES group	Hazard ratio (95%CI)	p value	$p_{interaction}$
Age, years						0·91
<75	2038	58/1017 (5·7%)	30/1021 (2·9%)	1·96 (1·26–3·05)	0·0027	
≥75	234	14/116 (12·3%)	8/118 (6·8%)	1·85 (0·78–4·42)	0·16	
Sex						0·57
Male	1574	49/769 (6·4%)	25/805 (3·1%)	2·08 (1·29–3·37)	0·0028	
Female	698	23/364 (6.3%)	13/334 (3.9%)	1·64 (0·83–3·24)	0·15	
Diabetes						0·97
Yes	620	21/282 (7·5%)	13/338 (3·9%)	1·97 (0·99–3·94)	0·054	
No	1652	51/851 (6·0%)	25/801 (3·1%)	1·94 (1·20–3·14)	0·0065	
Acute coronary syndrome						0·17
Yes	1257	44/630 (7·0%)	18/627 (2·9%)	2·49 (1·44–4·31)	0·0011	
No	1015	28/503 (5·6%)	20/512 (3·9%)	1·43 (0·80–2·53)	0·22	
Chronic kidney disease						0·35
Yes	203	12/95 (12·7%)	10/108 (9·3%)	1·37 (0·59–3·18)	0·46	
No	2069	60/1038 (5·8%)	28/1031 (2·7%)	2·16 (1·38–3·38)	0·0008	
Proximal LAD involvement						0·26
Yes	688	29/332 (8·8%)	12/356 (3·4%)	2·64 (1·35–5·17)	0·0047	
No	1584	43/801 (5·4%)	26/783 (3·3%)	1·63 (1·00–2·66)	0·048	
Small vessel coronary artery disease						0·020
Yes	1032	27/530 (5·1%)	22/502 (4·4%)	1·17 (0·67–2·05)	0·59	
No	1240	45/603 (7·5%)	16/637 (2·5%)	3·04 (1·72–5·38)	0·0001	
Bifurcation lesion						0·063
Yes	773	26/394 (6·7%)	20/379 (5·3%)	1·25 (0·70–2·23)	0·46	
No	1484	46/735 (6·3%)	18/749 (2·4%)	2·66 (1·54–4·59)	0·0004	
Multivessel treated						0·86
Yes	204	7/103 (6·8%)	4/101 (4·0%)	1·74 (0·51–5·93)	0·38	
No	2068	65/1030 (6·3%)	34/1038 (3·3%)	1·95 (1·29–2·96)	0·0016	
Centre volume						0·99
Low	1121	34/557 (6·1%)	18/564 (3·2%)	1·93 (1·09–3·41)	0·024	
High	1151	38/576 (6·6%)	20/575 (3·5%)	1·93 (1·13–3·32)	0·017	
Overall	**2272**	**72/1133 (6·4%)**	**38/1139 (3·4%)**	**1·93 (1·30–2·86)**	**0·0010**	

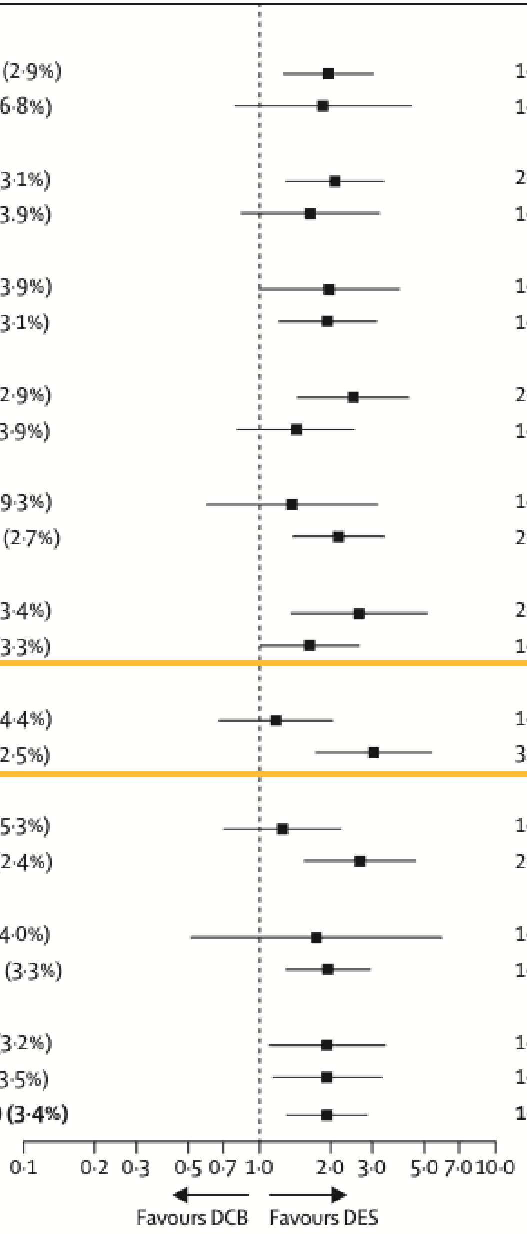
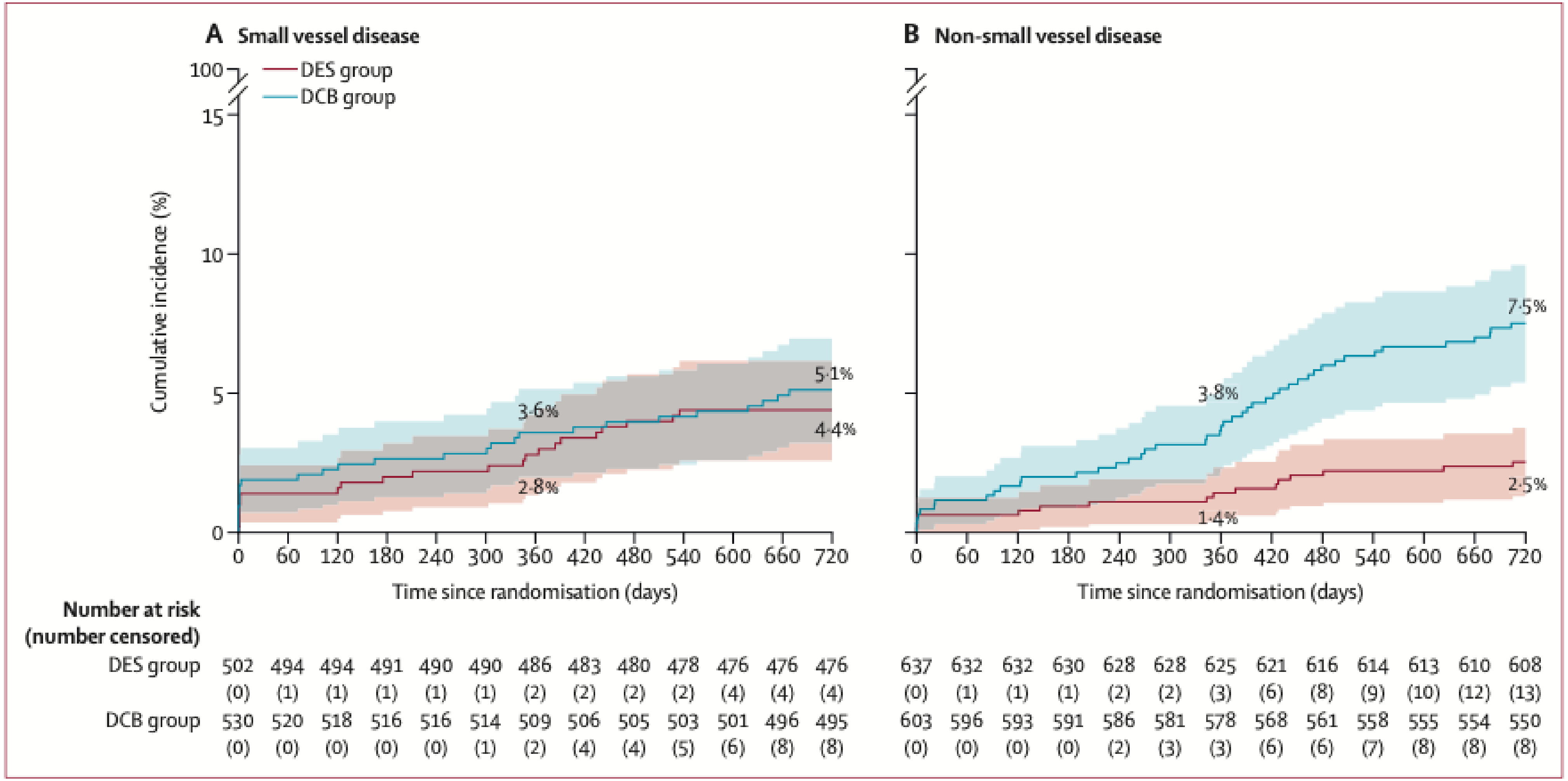


Figure 3: Subgroup analyses of the primary device-oriented composite endpoint at 24 months

Results


A Small vessel disease
B Non-small vessel disease
DES group
DCB group
Cumulative incidence (%)
100
15
10
5
0
3·6%
2·8%
5·1%
4·4%
3·8%
1·4%
7·5%
2·5%
0 60 120 180 240 300 360 420 480 540 600 660 720
Time since randomisation (days)
Number at risk
(number censored)
DES group 502 494 494 491 490 490 486 483 480 478 476 476 476
(0) (1) (1) (1) (1) (1) (2) (2) (2) (2) (4) (4) (4)
DCB group 530 520 518 516 516 514 509 506 505 503 501 496 495
(0) (0) (0) (0) (0) (1) (2) (4) (4) (5) (6) (8) (8)
DES group 637 632 632 630 628 628 625 621 616 614 613 610 608
(0) (1) (1) (1) (2) (2) (3) (6) (8) (9) (10) (12) (13)
DCB group 603 596 593 591 586 581 578 568 561 558 555 554 550
(0) (0) (0) (0) (2) (3) (3) (6) (6) (7) (8) (8) (8)

Discussion

- **OPUS-1 trial** (Lancet, 2000) demonstrated routine BMS implantation was better than POBA w/ rescue stenting strategy, mainly driven by increased risk of TVR
  - Though this study had substantial cross over with 37% of patients failing satisfactory results with POBA (then receiving rescue BMS), vs 21.7% in this study (thought to be driven by better lesion prep technologies, i.e., cutting or scoring balloons)
- The safety and efficacy of using DCBs in de novo lesions in coronary arteries irrespective of vessel diameter have been inferred; however, these conclusions are based on data with limited power, derived from small-scale angiographic investigations or non-randomized studies.
  - The BASKET-SMALL 2 trial, which was the largest randomized study investigating DCB to date (758 participants), showed that DCB with rescue DES was non-inferior to intended DES in patients with de novo small vessel disease, in terms of major adverse cardiovascular events over a period of 3 years
- The subgroup analysis in this trial also signals equipoise for treating small vessel disease with DCB vs DES

Discussion

- The incidence of target vessel revascularization (TVR) in the DES group was lower than what has been seen in previous studies
  - In the all-comers Global Leaders trial, the definition of non-complex coronary artery disease was similar to the current study and 68% of patients had non-complex PCI. At 2 years, the rate of TVR among these patients was 2.7%
  - In the PRECISE-DAPT pooled dataset (from eight randomized controlled trials), 79% of patients underwent non- complex PCI and the rate of TVR at 2 years was approximately 3.4%
  - In the current study, although the rates of TV-MI were not different between the two groups, the rates of TVR were 1.8% in the DES group and 4.8% in the DCB group

- The open-label design of the current study could have introduced biases because of the different thresholds for suspecting device failure and ordering stress testing or repeat invasive angiography in the DCB group, resulting in more frequent revascularization

Discussion

- **Limitations:**
  - open-label, but based on clinical endpoints (adjudicated by CEC)
  - secondary outcomes and subgroup analyses were not adjusted for multiple comparisons and should be interpreted as exploratory only
  - paclitaxel DCB vs sirolimus DES
  - while the Swide DCB has been demonstrated to be non-inferior to other DCB, caution should be taken assuming a class effect among other paclitaxel DCBs
  - only 30% female (underrepresented)
  - conducted in predominate east Asian population, limiting extrapolation to other ethnic or racial groups
  - DAPT duration was only mandated for 1 mo in DCB cohort
  - limited use of intravascular imaging (only ~10% of procedures)

AGENT IDE

Original Investigation

FREE

Cite Permissions Metrics

Paclitaxel-Coated Balloon vs Uncoated Balloon for Coronary In-Stent Restenosis
The AGENT IDE Randomized Clinical Trial

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JAMA
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Study Design / Methods

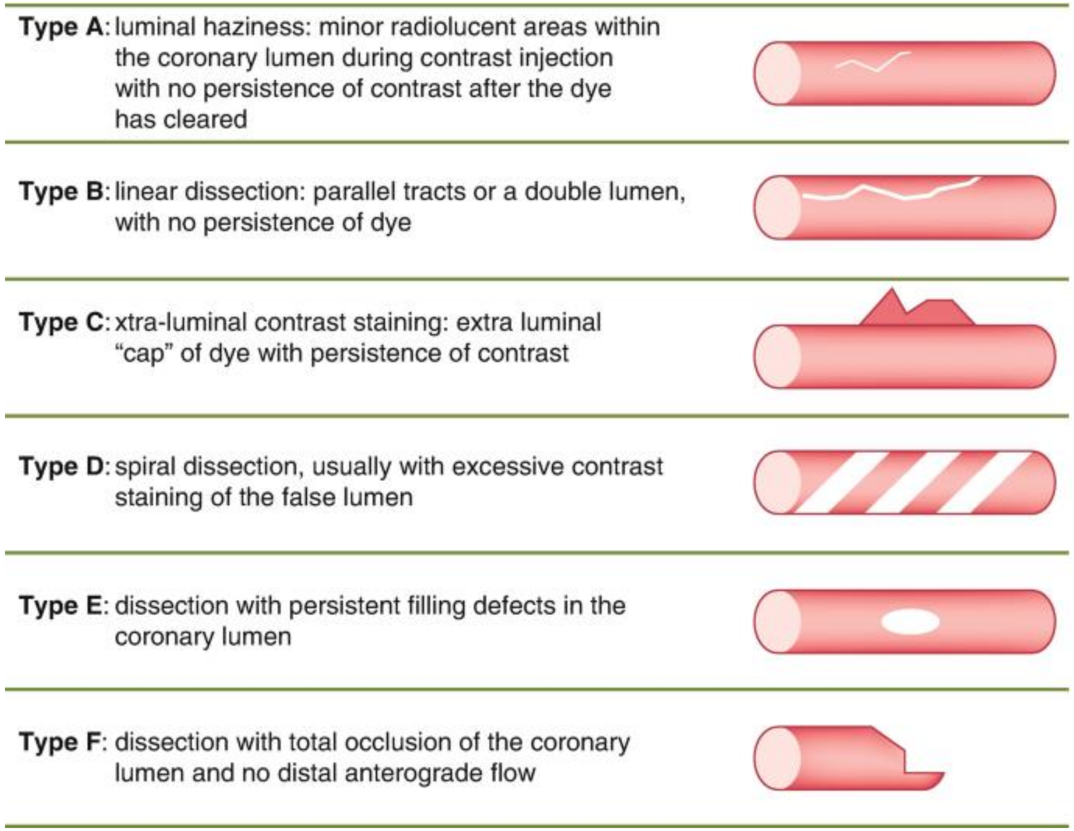
- Multicenter, single-blind, randomized controlled, superiority trial
  - enrolled 600 patients at 40 centers in US between May 2021 and August 2022
- compared angioplasty with DCB vs POBA for ISR
- AGENT paclitaxel-coated balloon (Boston Scientific Corp)

- **Inclusion Criteria:**
  - visually estimated reference vessel diameter 2.0 to 4.0 mm & length less than 26 mm
  - target lesion stenosis of <100% but >50% (in symptomatic patients) or >70% (in asymptomatic patients)

- **Key exclusion criteria:**
  - recent ST-elevation or Q-wave myocardial infarction
  - unprotected left main disease
  - saphenous vein or arterial graft disease
  - left ventricular ejection fraction less than 25%
  - thrombus in the target vessel

Study Design / Methods

- Patients who met selection criteria and underwent successful predilation of the target lesion were randomized in a 2:1 ratio for DCB vs POBA
  - Successful predilation defined as dilation with balloon catheter of appropriate length/diameter, or pretreatment with atherectomy, laser, or cutting/scoring balloon with no greater than 50% residual stenosis and no dissection greater than National Heart, Lung, and Blood Institute type C



Study Design / Methods

- Randomization was stratified by center and whether the target lesion had a single layer vs multiple stent layers of prior stents

- Only 1 target lesion was eligible for randomization for each patient

- A single nontarget lesion in a nontarget vessel could undergo PCI as determined by the operator prior to randomization

- Blinding:
  - Patients, core lab personnel, and the clinical events committee were blinded to assigned treatment; biostatistician remained blinded until analysis of the primary end point
  - Operators were <u>not</u> blinded due to different packaging of the paclitaxel-coated vs uncoated balloons

- DAPT (ASA + P2Y12i) for 1 month, followed by monotherapy for both arms

Study Design / Methods

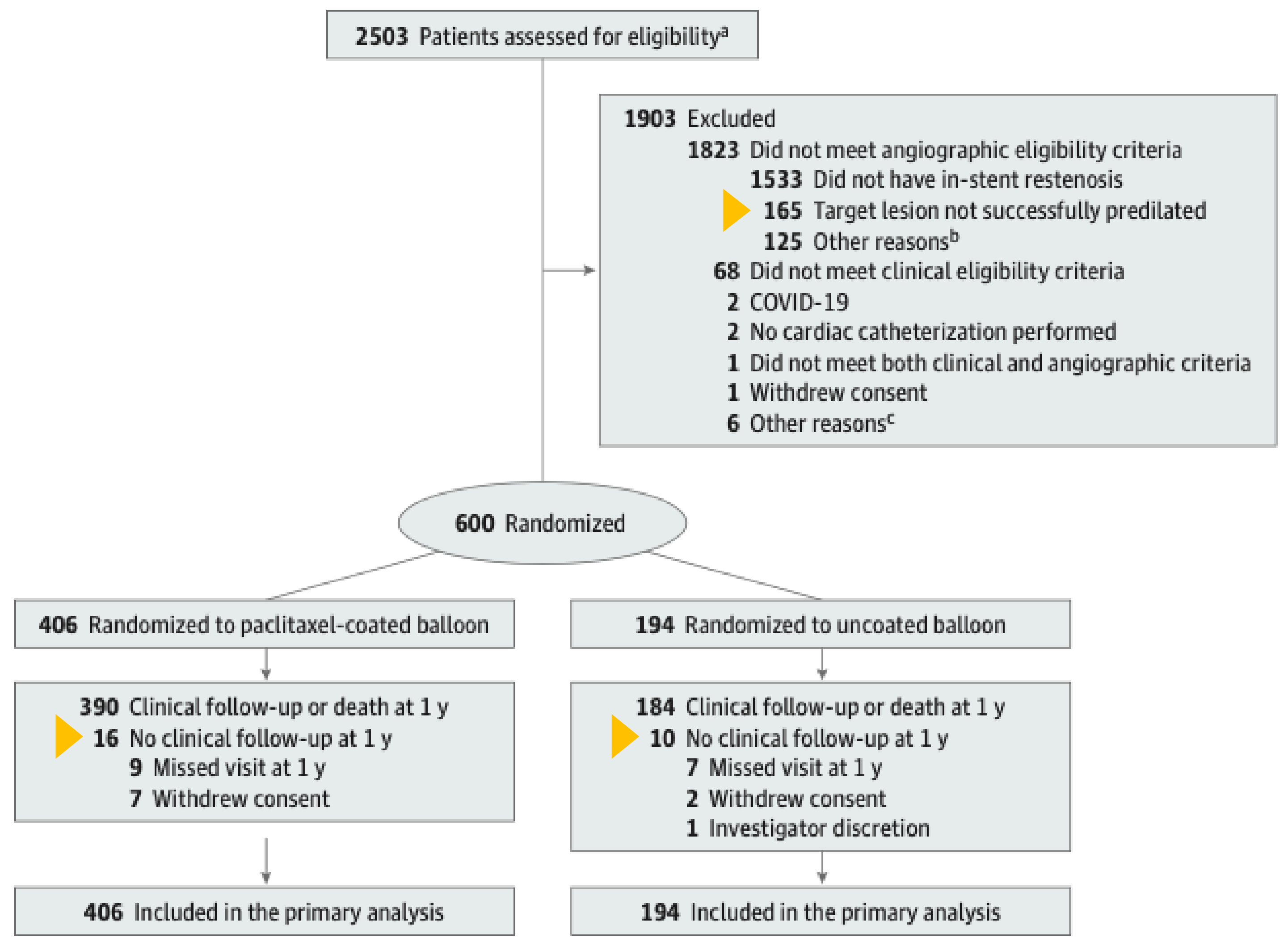
- Clinical follow-up was performed in hospital at 30 days, 6 months, and 12 months and will continue through 5 years

- **Primary end point:** rate of 12-month target lesion failure
  - composite of ischemia-driven target lesion revascularization, target vessel–related MI, or cardiac death
  - periprocedural MI occurring within 48 hrs was adjudicated by SCAI definition, and spontaneous MI occurring >48 hrs was adjudicated according to the fourth universal definition of MI

- **Secondary end points:**
  - individual components of the primary end point
  - all-cause mortality
  - stent thrombosis (as defined by the Academic Research Consortium)
  - periprocedural end points – rates of technical and procedural success

Study Design / Methods

Figure 1. Enrollment, Randomization, and Follow-Up in the AGENT IDE Trial



Study Design / Methods

Table 1. Baseline Clinical and Lesion Characteristics of the Patients[a]

Clinical characteristic	Paclitaxel-coated balloon (n = 406)	Uncoated balloon (n = 194)
Age, mean (SD), y	68.4 (9.8)	67.9 (9.7)
Sex		
Male	302 (74.4)	141 (72.7)
Female	104 (25.6)	53 (27.3)
Race and ethnicity[b]		
American Indian or Alaska Native	0	1 (0.5)
Asian	9 (2.2)	6 (3.1)
Black, of African heritage	32 (7.9)	10 (5.2)
Hispanic or Latino	26 (6.4)	9 (4.6)
Native Hawaiian or Other Pacific Islander	1 (0.2)	1 (0.5)
White	304 (74.9)	148 (76.3)
Not disclosed	23 (5.7)	18 (9.3)
Other[c]	16 (3.9)	3 (1.5)
Body mass index, mean (SD) [No.][d]	30.0 (5.5) [396]	30.1 (5.9) [191]
Currently smokes	42/406 (10.3)	19/194 (9.8)
Medical history		
Hypertension	383/405 (94.6)	186/194 (95.9)
Hyperlipidemia	382/404 (94.6)	184/194 (94.8)
Multivessel coronary artery disease	317/400 (79.3)	149/190 (78.4)
Current diabetes	206/404 (51.0)	97/194 (50.0)
Myocardial infarction	198/398 (49.7)	95/190 (50.0)
Coronary artery bypass graft surgery	124/403 (30.8)	55/192 (28.6)
Insulin-treated diabetes	93/404 (23.0)	49/194 (25.3)
Congestive heart failure	92/401 (22.9)	41/192 (21.4)
Left main coronary artery disease	89/394 (22.6)	39/188 (20.7)
Peripheral vascular disease	78/401 (19.5)	31/192 (16.1)
Kidney disease	74/402 (18.4)	32/193 (16.6)
Transient ischemic attack or cerebrovascular accident	57/402 (14.2)	19/194 (9.8)
Indication for procedure		
Stable angina	225 (55.4)	100 (51.5)
Non-ST-elevation acute coronary syndrome	149 (36.7)	73 (37.6)
Silent ischemia	7 (1.7)	5 (2.6)
Other indication[e]	25 (6.2)	16 (8.2)
Lesion characteristics[f]		
Stent layer in target lesion[g]		
Single	230 (56.7)	112 (57.7)
Multiple	176 (43.3)	82 (42.3)
Target lesion vessel		
Right coronary artery	156 (38.4)	69 (35.6)
Left anterior descending artery	141 (34.7)	69 (35.6)
Left circumflex artery	98 (24.1)	47 (24.2)
Left main coronary artery	11 (2.7)	9 (4.6)
Lesion length, mean (SD), mm [No.]	12.8 (6.3) [404]	11.8 (6.6) [188]
Reference vessel diameter, mean (SD), mm [No.]	2.7 (0.5) [405]	2.7 (0.5) [192]
In-lesion minimum lumen diameter, mean (SD), mm [No.]	1.0 (0.4) [405]	0.9 (0.4) [191]
In-lesion diameter stenosis, mean (SD), % [No.]	65.0 (12.1) [405]	66.4 (12.8) [191]
Mehran in-stent restenosis pattern[h]		
1A (Articulation)	0/403	0/189
1B (Margin)	4/403 (1.0)	2/189 (1.1)
1C (Focal)	148/403 (36.7)	84/189 (44.4)
1D (Multifocal)	3/403 (0.7)	2/189 (1.1)
2 (Diffuse intrastent)	230/403 (57.1)	89/189 (47.1)
3 (Diffuse proliferative)	16/403 (4.0)	9/189 (4.8)
4 (Total occlusion)	2/403 (0.5)	3/189 (1.6)
Thrombolysis In Myocardial Infarction grade flow		
0 (Complete occlusion)	8/405 (2.0)	6/192 (3.1)
1 (Some flow)	2/405 (0.5)	3/192 (1.6)
2 (Partial flow)	21/405 (5.2)	8/192 (4.2)
3 (Normal flow)	374/405 (92.3)	175/192 (91.1)

Statistical Analysis

- This study used an adaptive group sequential design with an initial planned enrollment of 480 patients and 1 formal interim analysis on the 1-year data after randomization of the first 90% of patients, with the expectation that the first 40% of randomized patients would have 1-year follow-up at that time (details provided in Supplement 1).
  - Due to rapid enrollment in the trial, no patients had completed 1-year follow-up at the time of the planned interim analysis, and therefore the data and safety monitoring committee recommended to continue enrollment to the maximum of 600 patients.

- With a sample size of 480 patients, the study had 85% power to demonstrate superiority (1-sided $P < .025$) of DCB vs POBA, assuming respective event rates of 10.6% and 21.2%, with an expected attrition rate of 3%.
  - Given uncertainty in the accuracy of these estimated event rates, the interim analysis was prespecified to be performed by the data and safety monitoring committee for potential sample size re-estimation of up to 600 patients.

- The primary analysis was performed as **intention-to-treat**
- Continuous variables are presented as means (SDs) and differences between groups tested with the *t* test.
- Discrete variables are reported as percentages and counts, and differences between groups were assessed by the $\chi 2$ or Fisher exact test, as appropriate. A *z* test with unpooled variance for the difference of 2 proportions was used to test the primary end point hypothesis of superiority of paclitaxel-coated balloon over the uncoated balloon. Differences in the rates of the primary end point and additional prespecified outcomes are reported using survival methodology as cumulative incidence rates and associated hazard ratios (HRs) with 95% CIs.
- Prespecified subgroup analyses of the primary outcome were analyzed using Cox regression models that included subgroup by treatment group interaction terms

Results

Figure 2. One-Year Time-to-Event Curves for the Primary End Point and Individual Components

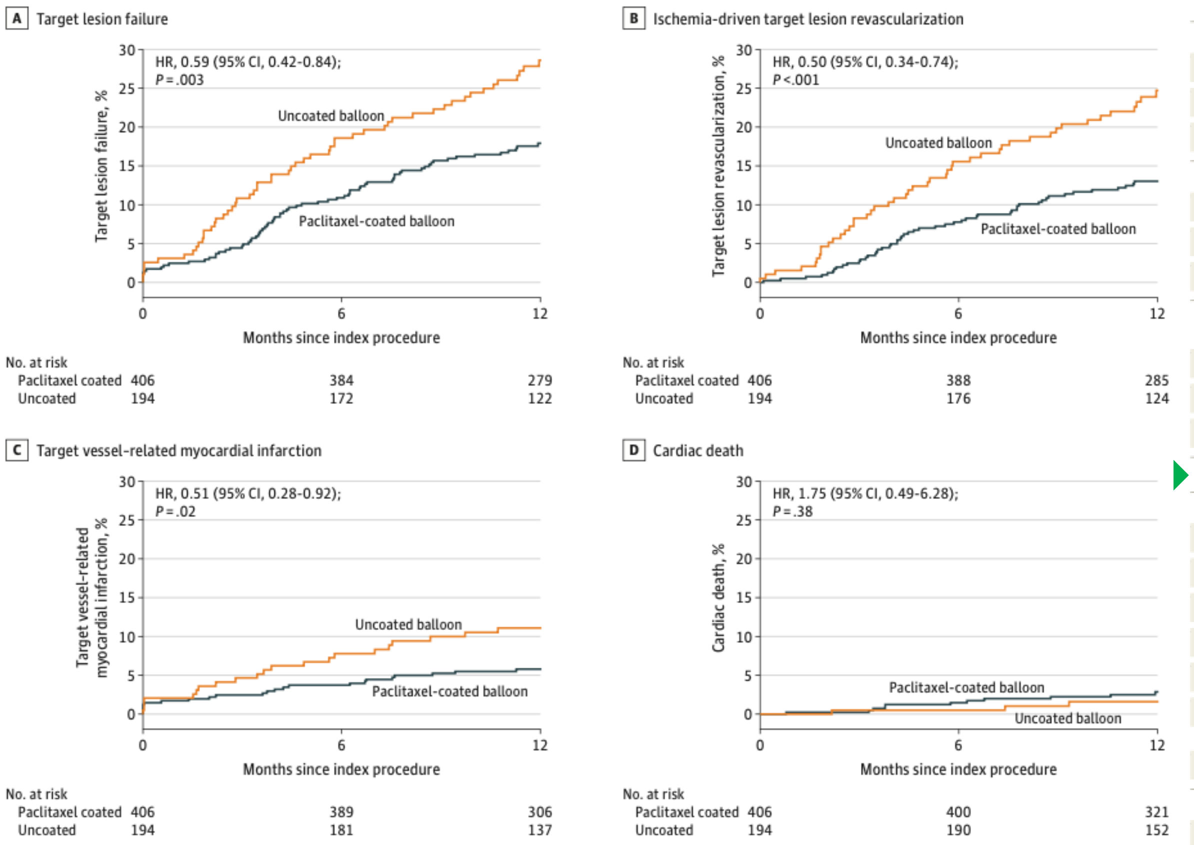
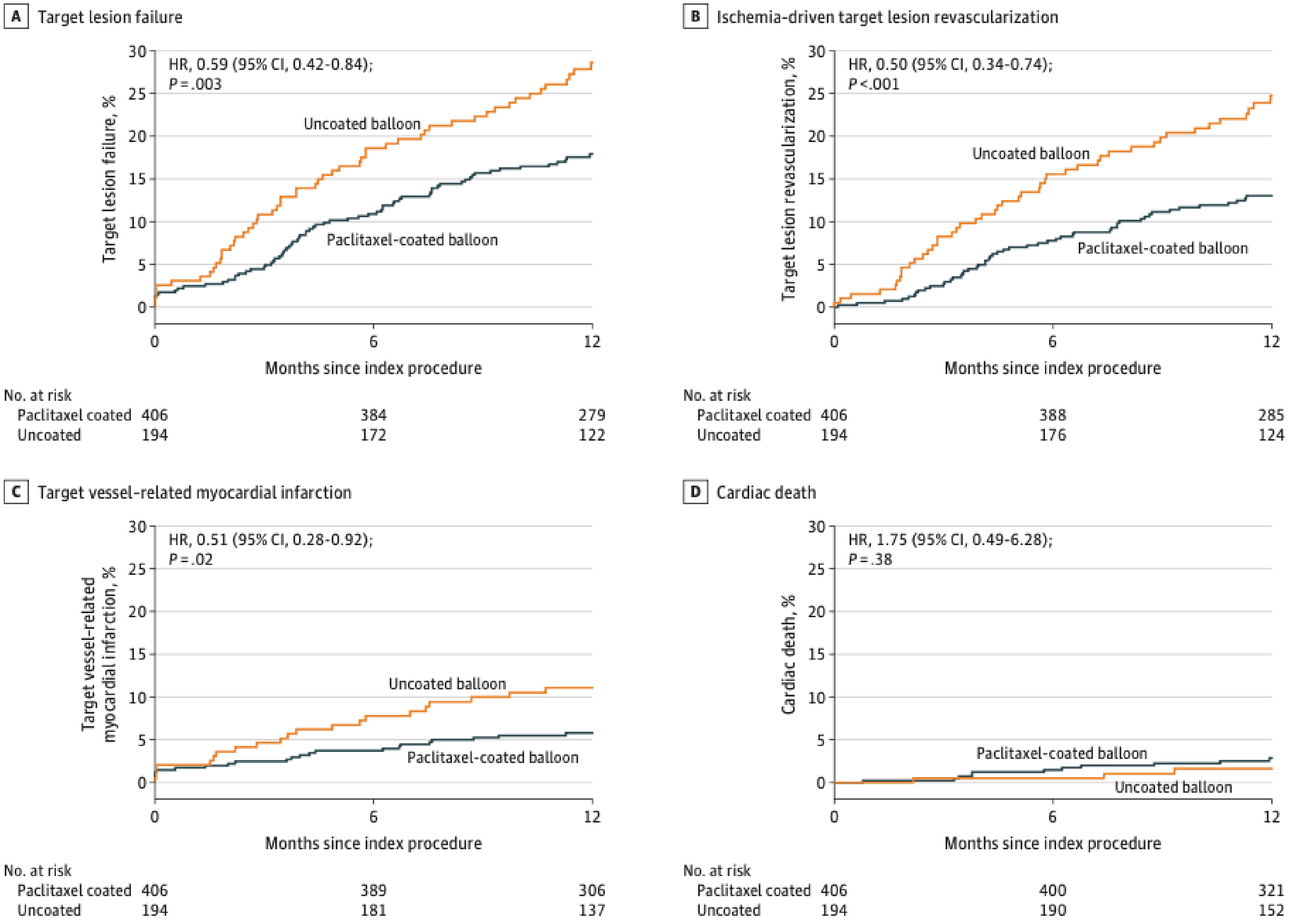


Table 2. Prespecified Clinical Outcomes Through 1 Year

Clinical outcomes[a]	No. % Paclitaxel-coated balloon (n = 406)	Uncoated balloon (n = 194)	Difference (95% CI), %	HR (95% CI)	Log-rank *P* value
Primary outcome					
Target lesion failure[b]	71 (17.9)	54 (28.6)	−10.7 (−18.2 to −3.2)	0.59 (0.42 to 0.84)	.003
Additional prespecified outcomes					
Target lesion revascularization[c]					
Overall	51 (13.0)	46 (24.7)	−11.6 (−18.7 to −4.5)	0.50 (0.34 to 0.74)	.001
PCI	39 (10.0)	43 (22.8)	−12.9 (−19.6 to −6.1)	0.41 (0.26 to 0.63)	<.001
CABG	13 (3.3)	8 (4.5)	−1.2 (−4.8 to 2.4)	0.77 (0.32 to 1.86)	.56
Target vessel revascularization[c]					
Overall	56 (14.3)	49 (26.2)	−11.8 (−19.1 to −4.6)	0.51 (0.35 to 0.75)	.001
PCI	44 (11.3)	46 (24.4)	−13.1 (−20.0 to −6.2)	0.43 (0.28 to 0.64)	<.001
CABG	14 (3.6)	8 (4.5)	−0.9 (−4.5 to 2.7)	0.83 (0.35 to 1.98)	.68
Target vessel revascularization not in target lesion[c,d]					
Overall	20 (5.1)	13 (6.9)	−1.8 (−6.0 to 2.4)	0.72 (0.36 to 1.46)	.36
PCI	17 (4.3)	11 (5.8)	−1.5 (−5.3 to 2.4)	0.73 (0.34 to 1.55)	.41
CABG	4 (1.0)	3 (1.6)	−0.6 (−2.6 to 1.5)	0.64 (0.14 to 2.85)	.55
Target vessel failure[e]	73 (18.4)	57 (30.1)	−11.7 (−19.3 to −4.1)	0.57 (0.40 to 0.81)	.001
Myocardial infarction	29 (7.5)	23 (12.1)	−4.7 (−10.0 to 0.7)	0.58 (0.34 to 1.01)	.05
Related to target vessel	23 (5.8)	21 (11.1)	−5.3 (−10.3 to −0.2)	0.51 (0.28 to 0.92)	.02
Q-wave myocardial infarction	1 (0.3)	1 (0.5)	−0.3 (−1.4 to 0.9)	0.48 (0.03 to 7.64)	.59
Related to target vessel	0	1 (0.5)	−0.5 (−1.5 to 0.5)	NA	.15
Non-Q-wave myocardial infarction	28 (7.2)	22 (11.6)	−4.4 (−9.7 to 0.9)	0.59 (0.34 to 1.03)	.06
Related to target vessel	23 (5.8)	20 (10.6)	−4.8 (−9.7 to 0.2)	0.54 (0.30 to 0.98)	.04
Related to target vessel-unknown[f]	11 (2.8)	4 (2.2)	0.7 (−2.0 to 3.3)	1.32 (0.42 to 4.16)	.63
Not related to target vessel	5 (1.4)	2 (1.1)	0.4 (−1.5 to 2.3)	1.19 (0.23 to 6.12)	.84
All death[g]	16 (4.1)	7 (3.7)	0.4 (−3.0 to 3.7)	1.09 (0.45 to 2.65)	.85
Cardiac	11 (2.9)	3 (1.6)	1.3 (−1.2 to 3.7)	1.75 (0.49 to 6.28)	.38
Noncardiac	5 (1.3)	4 (2.2)	−0.9 (−3.3 to 1.5)	0.60 (0.16 to 2.22)	.43
Stent thrombosis[g]	1 (0.4)	7 (3.7)	−3.3 (−6.1 to −0.5)	0.07 (0.01 to 0.55)	.001
Definite or probable	0	6 (3.2)	−3.2 (−5.7 to −0.7)	NA	<.001
Definite	0	6 (3.2)	−3.2 (−5.7 to −0.7)	NA	<.001
Probable	0	0	0.0 (0.0 to 0.0)	NA	Undefined
Possible	1 (0.4)	1 (0.5)	−0.2 (−1.4 to 1.1)	0.48 (0.03 to 7.63)	.59

Results

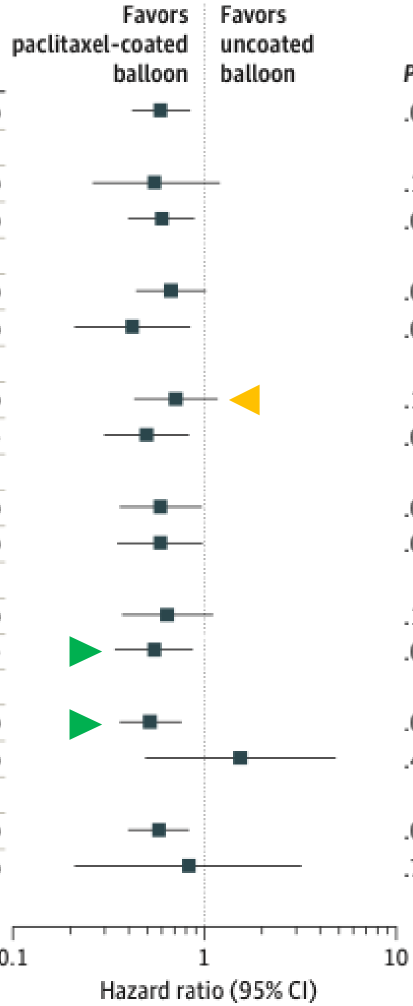
Figure 2. One-Year Time-to-Event Curves for the Primary End Point and Individual Components



Results

Figure 3. Prespecified Subgroup Analyses of the Primary Outcome at 1 Year

Subgroup	No. (%) Paclitaxel-coated balloon (n=406)	No. (%) Uncoated balloon (n=194)	Difference (95% CI), %	Hazard ratio (95% CI)	P value	P value for interaction
All patients (N=600)	71 (17.9)	54 (28.6)	-10.7 (-18.2 to -3.2)	0.59 (0.42-0.84)	.003	
Sex						
Female (n=157)	14 (14.3)	12 (23.1)	-8.8 (-22.2 to 4.6)	0.55 (0.26-1.20)	.13	.84
Male (n=443)	57 (19.1)	42 (30.7)	-11.6 (-20.6 to -2.6)	0.60 (0.40-0.89)	.01	
Age, y						
<75 (n=436)	55 (19.0)	37 (27.7)	-8.6 (-17.5 to 0.3)	0.67 (0.44-1.02)	.06	.26
≥75 (n=164)	16 (14.9)	17 (31.0)	-16.1 (-30.0 to -2.1)	0.42 (0.21-0.84)	.01	
Diabetes						
Yes[a] (n=269)	38 (21.6)	26 (29.2)	-7.6 (-18.9 to 3.6)	0.71 (0.43-1.17)	.18	.35
No[b] (n=329)	33 (15.2)	28 (28.0)	-12.8 (-22.9 to -2.7)	0.50 (0.30-0.83)	.006	
Vessel size[c]						
Small (RVD < 2.75 mm) (n=332)	39 (17.3)	27 (27.4)	-10.1 (-20.2 to 0.0)	0.59 (0.36-0.97)	.04	.98
Large (RVD=2.75 mm) (n=265)	32 (18.9)	27 (30.5)	-11.7 (-23.0 to -0.3)	0.59 (0.35-0.98)	.04	
Stent layer[d]						
Single (n=341)	30 (13.5)	22 (20.2)	-6.7 (-15.5 to 2.1)	0.64 (0.37-1.11)	.11	.66
Multiple (n=258)	41 (23.8)	32 (40.0)	-16.2 (-28.9 to -3.6)	0.55 (0.34-0.87)	.009	
Treatment[e]						
Only target lesion (n=523)	60 (17.4)	50 (30.5)	-13.2 (-21.3 to -5.0)	0.52 (0.36-0.76)	.001	.08
Target and nontarget lesion (n=77)	11 (21.9)	4 (16.3)	5.6 (-12.9 to 24.2)	1.54 (0.49-4.83)	.46	
History of treatment						
Drug-eluting stent only (n=527)	64 (18.5)	50 (30.1)	-0.55 (-0.92 to -0.18)	0.58 (0.40-0.83)	.003	.60
Any bare metal stent (n=57)	7 (16.8)	3 (20.0)	-3.2 (-26.4 to 20.0)	0.83 (0.21-3.21)	.79	



Discussion

- Limitations:
  - operators were not blinded, conducted at high-volume sites with higher rates IVUS, does not address DCB vs DES, not powered to assess important subgroups or low-frequency events (IST)
- Largest RCT to date examining ISR PCI with DCB vs POBA and first conducted in US
  - Approved and utilized more broadly internationally, based on smaller RCT
- DCB demonstrated lower rate of target lesion failure
  - Largely driven by lower rates of ischemia-driven revascularization and target vessel MI
  - rate of target vessel myocardial infarction at 1 year was 5.8% in the paclitaxel-coated balloon group compared with 11.1% in the uncoated balloon group, and no cases of definite or probable stent thrombosis occurred among the paclitaxel- coated balloon group compared with 6 cases (3.2%) within the uncoated balloon group
- ISR rates of 20-40% in BMS and still represent ~10% of cases of PCI annually

JAMA

QUESTION Is treatment with a coronary paclitaxel-coated balloon superior to an uncoated balloon for 1-year target lesion failure in patients undergoing percutaneous coronary intervention for in-stent restenosis?

CONCLUSION This clinical trial found that treatment with a paclitaxel-coated balloon offers a potentially beneficial treatment strategy for the management of coronary in-stent restenosis.

POPULATION

443 Men
157 Women

Adults with in-stent restenosis (lesion length <26 mm and reference vessel diameter >2.0 mm to ≤4.0 mm)

Mean age: 68 years

LOCATIONS

40 Centers in the US

INTERVENTION

600 Patients randomized

406 **Paclitaxel-coated balloon**
Coronary angioplasty with a paclitaxel-coated balloon

194 **Uncoated balloon**
Coronary angioplasty with an uncoated balloon

PRIMARY OUTCOME

1-year target lesion failure, defined as the composite of ischemia-driven target lesion revascularization, target vessel–related myocardial infarction, or cardiac death

FINDINGS

Target lesion failure

Paclitaxel-coated balloon
71 of 406 patients

Uncoated balloon
54 of 194 patients

Target lesion failure was significantly lower in the paclitaxel-coated balloon group:

Between-group difference, −10.7% (95% CI, −18.2% to −3.2%)

Hazard ratio, 0.59 (95% CI, 0.42 to 0.84)

Yeh RW, Shlofmitz R, Moses J, et al; AGENT IDE Investigators. Paclitaxel-coated balloon vs uncoated balloon for coronary in-stent restenosis: the AGENT IDE randomized clinical trial. *JAMA*. Published March 9, 2024. doi:10.1001/jama.2024.1361

AMERICAN COLLEGE of CARDIOLOGY

REC-CAGEFREE I

Rescue Stenting vs. Intended Stenting in Patients With De Novo Coronary Artery Lesions

OBJECTIVE

Assess the noninferiority of drug-coated balloon (DCB) angioplasty with rescue stenting compared with intended stenting with second-generation thin-strut sirolimus-eluting stents (DES) in patients with de novo coronary artery lesions.

STUDY METHODS

TOTAL NO. OF PATIENTS: 2,272
(43 sites in China; median age 62 years; 30.7% Female)

INCLUSION CRITERIA:
De novo, non-complex coronary artery disease and an indication for PCI

STUDY DESIGN:
Randomized 1:1 to DCB with rescue stenting or intended stenting with DES

PRIMARY ENDPOINTS

AT 24 MONTHS, FEWER PATIENTS IN THE DCB GROUP VS. THE DES GROUP EXPERIENCED A DEVICE-ORIENTED COMPOSITE ENDPOINT (CARDIOVASCULAR DEATH, TARGET VESSEL MYOCARDIAL INFARCTION, AND CLINICALLY AND PHYSIOLOGICALLY INDICATED TARGET LESION REVASCULARIZATION):

DCB GROUP: 72 PATIENTS **DES GROUP: 38 PATIENTS**

NONINFERIORITY WAS NOT MET.

CONCLUSION

DES should remain the preferred treatment for patients with de novo coronary artery lesions.

Key Learning Points & Considerations

- Comparing modalities of PCI (i.e., DCB vs POBA vs DES) requires consideration of clinical scenario (i.e., ACS vs stable ischemic heart disease vs ISR) as well as target lesion features (e.g., vessel caliber, length, or complex features)

- Currently, DCB have not shown clinical equipoise to DES for routine treatment of de novo lesions
  - though multiple trials have demonstrated or at least signaled equipoise for treatment of de novo lesions in small caliber vessels

- DCB > POBA for ISR
  - though DES is still considered the gold standard therapy for single layered lesions

- Future generations of DCB technologies may redefine role for primary PCI

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